



Management Science

Publication details, including instructions for authors and subscription information:
<http://pubsonline.informs.org>

Tick Size and Earnings Guidance in Small-Cap Firms: Evidence from the SEC's Tick Size Pilot Program

Yangyang Chen, Jeffrey Ng, Emmanuel Ofosu, Xin Yang

To cite this article:

Yangyang Chen, Jeffrey Ng, Emmanuel Ofosu, Xin Yang (2024) Tick Size and Earnings Guidance in Small-Cap Firms: Evidence from the SEC's Tick Size Pilot Program. *Management Science* 70(8):5386–5407. <https://doi.org/10.1287/mnsc.2023.4930>

Full terms and conditions of use: <https://pubsonline.informs.org/Publications/Librarians-Portal/PubsOnLine-Terms-and-Conditions>

This article may be used only for the purposes of research, teaching, and/or private study. Commercial use or systematic downloading (by robots or other automatic processes) is prohibited without explicit Publisher approval, unless otherwise noted. For more information, contact permissions@informs.org.

The Publisher does not warrant or guarantee the article's accuracy, completeness, merchantability, fitness for a particular purpose, or non-infringement. Descriptions of, or references to, products or publications, or inclusion of an advertisement in this article, neither constitutes nor implies a guarantee, endorsement, or support of claims made of that product, publication, or service.

Copyright © 2023, INFORMS

Please scroll down for article—it is on subsequent pages



With 12,500 members from nearly 90 countries, INFORMS is the largest international association of operations research (O.R.) and analytics professionals and students. INFORMS provides unique networking and learning opportunities for individual professionals, and organizations of all types and sizes, to better understand and use O.R. and analytics tools and methods to transform strategic visions and achieve better outcomes. For more information on INFORMS, its publications, membership, or meetings visit <http://www.informs.org>

Tick Size and Earnings Guidance in Small-Cap Firms: Evidence from the SEC's Tick Size Pilot Program

Yangyang Chen,^a Jeffrey Ng,^b Emmanuel Ofosu,^a Xin Yang^{c,*}

^aCity University of Hong Kong, Kowloon, Hong Kong, China; ^bThe University of Hong Kong, Pok Fu Lam, Hong Kong, China; ^cCentral University of Finance and Economics, Beijing 102206, China

*Corresponding author

Contact: ychen722@cityu.edu.hk,  <https://orcid.org/0000-0003-3665-8261> (YC); jeffngty@hku.hk,  <https://orcid.org/0000-0003-1430-3646> (JN); eofosu2-c@my.cityu.edu.hk (EO); xinyang515@outlook.com (XY)

Received: July 19, 2021

Revised: December 4, 2022;
February 17, 2023

Accepted: March 7, 2023

Published Online in Articles in Advance:
September 27, 2023

<https://doi.org/10.1287/mnsc.2023.4930>

Copyright: © 2023 INFORMS

Abstract. The Securities and Exchange Commission's 2016 Tick Size Pilot Program was a natural experiment that imposed increases in tick size for randomly selected small-cap firms. Using a difference-in-differences research design, we examine the effect of this increase in tick size on earnings guidance. We find that after initiation of the program, treatment firms provide significantly less earnings guidance. We provide further evidence that this decrease is driven by increases in investors' fundamental information acquisition and in firms' financial reporting quality, consistent with firms reducing earnings guidance when investors are already more informed. The decrease is stronger for firms with higher proprietary costs of disclosure, consistent with firms being more likely to reduce costly disclosure when investors are more informed. In contrast, the decrease is weaker for firms with greater external financing needs, consistent with these firms continuing to seek the benefits of disclosure, even when investors are more informed. Taken together, our results suggest that an increase in tick size makes investors more informed, which, in turn, reduces the need for firms to provide earnings guidance, though the extent of the reduction depends on the costs and benefits of providing earnings guidance.

History: Accepted by Suraj Srinivasan, accounting.

Supplemental Material: The data files are available at <https://doi.org/10.1287/mnsc.2023.4930>.

Keywords: Tick Size Pilot Program • investor information acquisition • earnings guidance

1. Introduction

A cornerstone of efficient capital markets is investors' pursuit and acquisition of information. An extensive body of theoretical and empirical literature has considered firm disclosure in the presence of information asymmetry, investor attention, and information-processing incentives and capabilities (Healy and Palepu 2001, Verrecchia 2001, Blankespoor et al. 2020). Other literature has examined capital market outcomes of firms' information environments. For example, Collier and Yohn (1997) document that in the short (long) run, the provision of earnings guidance increases (decreases) bid-ask spreads in the equity market. Beyer et al. (2010) estimate that earnings guidance better explains total stock return variance, compared with mandatory earnings reports and Securities and Exchange Commission (SEC) filings. Some studies examine how capital market conditions can affect firms' information environments. For example, recent research has used the SEC's 2016 Tick Size Pilot Program (TSPP), which increased tick size for randomly selected small-cap firms, as a natural experiment to show how the increase incentivizes investors to engage more in fundamental information acquisition. In

particular, Lee and Watts (2021) document that increasing a firm's tick size encourages its investors to acquire fundamental information, such as SEC filings. Ahmed et al. (2020) and Ye et al. (2023) build on Lee and Watts (2021) and document that increased tick size causes an increase in financial reporting quality and managerial learning from stock prices. They attribute their findings to investors becoming more informed via increased fundamental information acquisition. In our study, we extend this research by examining the link between increased tick size and provision of earnings guidance through the lens of investors becoming more informed.¹

We argue that a study of the above link is important for at least two reasons. The first reason relates to the regulatory objective of the TSPP. This data-driven program provided a way for regulators to evaluate whether widening the tick size for securities of smaller-cap companies would impact the trading, liquidity, and market quality of those securities. As voluntary disclosures, such as earnings guidance, can affect these market conditions, it is important to study how an increase in tick size can affect voluntary disclosure to gain a holistic understanding of how tick size affects market conditions. Second,

using the TSPP as an exogenous shock to capital market participants' information set enables us to deepen our understanding of the link between this information and corporate voluntary disclosure. For example, Balakrishnan et al. (2014) use brokerage firm closures to study how information loss affects investors and find that firms substitute for this loss by providing more earnings guidance. An increase in tick size can be viewed as a setting in which investors are incentivized to gather fundamental information and are thus more informed (Lee and Watts 2021). As such, this paper complements the findings of Balakrishnan et al. (2014) and related literature by using an economically relevant shock to study the capital market participants' information set and management earnings guidance nexus.

In brief, the arguments linking an increase in tick size to reduced earnings guidance are as follows. First, Lee and Watts (2021) find that an increase in tick size increases fundamental information acquisition by investors about the treatment firms. Fundamental information acquisition increases the overall information contained in stock prices, and more informative stock prices, in turn, lower information asymmetry between firms and their investors (Rindi and Werner 2019, Chung et al. 2020, Lee and Watts 2021). However, managers have fewer incentives to engage in the potentially costly provision of earnings guidance if investors are already informed. Second, Ahmed et al. (2020) find that an increase in tick size improves the financial reporting quality of treatment firms and attribute this finding to enhanced investor scrutiny of firms' financial reporting choices. To the extent that increased financial reporting quality helps investors better forecast future earnings, investors have less need for voluntary disclosure, which reduces managers' incentives to provide earnings guidance. Third, to the extent that investor scrutiny of firms' disclosure increases in general, managers might be more constrained in issuing misleading earnings guidance to achieve objectives such as meeting or beating analyst forecasts, which, in turn, can reduce the issuance of earnings guidance.

Conversely, increased tick size could encourage managers to increase earnings guidance as a way to improve investors' portfolio analyses. If increased tick size motivates investors to become more informed, investor demand for analysis-enhancing information, such as earnings guidance, might increase (Rogers and Stocken 2005, Kothari et al. 2009, Hutton et al. 2012, Rogers and Van Buskirk 2013). Furthermore, to benefit from market feedback when there is more information processing by investors, managers can strategically issue guidance, observe the market response, and then use the information to guide their corporate decision making (Bond et al. 2012, Gao and Liang 2013, Jayaraman and Wu 2020). Therefore, how an increase in tick size affects managers' earnings guidance is ultimately an empirical question.

The TSPP targets small-cap firms, whose stock trading, liquidity, and market quality tend to be more affected by tick size (Lee and Watts 2021). We believe this setting is a powerful one to study the strategic altering of earnings guidance in response to an increase in tick size, with documented important informational effects, such as increased fundamental information acquisition (Lee and Watts 2021), improved financial reporting quality (Ahmed et al. 2020), and enhanced managerial learning for stock prices (Ye et al. 2023). To the extent that smaller firms are more cautious in providing earnings guidance (e.g., Cox 1985, Chow and Wong-Boren 1987, Kasznik and Lev 1995), possibly because their costs of doing so are higher, their disclosure decisions also might be more sensitive to changes in the information environment (e.g., changes in information asymmetries and demand for voluntary disclosure).

Following the aforementioned studies on the TSPP, we use a difference-in-differences research design to investigate the effect of an increase in tick size on earnings guidance. We find that, relative to the control firms, the treatment firms experience significant decreases in the likelihood and frequency of earnings guidance after initiation of the TSPP. The effect of the increased tick size on earnings guidance is also economically significant. Specifically, we find that, compared with control firms, treatment firms experience 10.8% and 10.4% decreases in forecast likelihood and forecast frequency, relative to the respective sample means, after TSPP initiation. Our results indicate that an increase in tick size leads to a reduction in earnings guidance. Our results are robust to parallel-trend tests, in that we find no pre-existing trends, and treatment firms significantly reduce their earnings guidance starting in the second quarter after TSPP initiation. We further perform several robustness checks using alternative event windows and a propensity-score-matched sample, and our results hold. We also perform two placebo tests and find that a pseudo-event does not lead treatment firms to decrease earnings guidance and that pseudo-treatment firms do not experience decreased earnings guidance after TSPP initiation. Collectively, these results suggest that the impact of an increased tick size on earnings guidance is likely causal.

Next, we run some channel tests to examine the three arguments that we propose as explanations for a negative effect of an increase in tick size on earnings guidance. First, consistent with our argument that an increased tick size increases investors' fundamental information acquisition, which, in turn, reduces managers' need to provide earnings guidance to already-informed investors, we find that the negative impact of the TSPP on earnings guidance is concentrated among treatment firms experiencing increased downloads of their SEC filings. Second, consistent with improved financial reporting quality enabling investors to better forecast future earnings, thereby reducing the need for earnings guidance, we find that the negative impact of the TSPP on earnings guidance is

concentrated among treatment firms with improved reporting quality. Third, we examine the argument that the reduction in earnings guidance could be due to constraints in issuing misleading earnings guidance. We do not find evidence that the impact of the TSPP on earnings guidance is concentrated among treatment firms with incentives to use earnings guidance to opportunistically manage investors' expectations. Overall, the tests indicate that when investors become more informed, either because of greater information acquisition or availability of higher-quality financial reports, managers have fewer incentives to provide potentially costly earnings guidance.

We then run two cross-sectional analyses to further examine the above notion that increased tick size reduces managers' incentives to provide potentially costly earnings guidance. First, we find evidence that the negative effect of the TSPP on earnings guidance is more pronounced for firms with higher proprietary disclosure costs. We argue that firms with higher disclosure costs are more likely to reduce earnings guidance when the benefits of disclosure decrease, such as when investors become more informed due to an increase in tick size. Second, we find that the negative impact of the TSPP on earnings guidance is less pronounced for firms in need of external financing, as these firms are likely to continue seeking the benefits of disclosure.

Our study makes several contributions to the literature. First, we contribute to research examining how managers adjust corporate policies in response to trading conditions in secondary markets (e.g., Massa et al. 2015, Zuo 2016, Ahmed et al. 2020). Within the context of an increase in tick size, Ahmed et al. (2020) study how increased tick size affects financial reporting quality. Our paper complements and extends Ahmed et al. (2020), in that they study the effect on an earnings-related mandatory disclosure, whereas we investigate the effect on an earnings-related voluntary disclosure. A key insight of our paper is that when fundamental investors are incentivized to gather fundamental information and when higher-quality financial reports are available to them, managers feel less need to assist investors and, therefore, offer less earnings guidance. Relatedly, our paper can be viewed as one that extends the literature on the substitutive effect between firms' disclosure and capital market participants' information set. In addition to the aforementioned study on brokerage firm closures by Balakrishnan et al. (2014), studies by Koch et al. (2013) and Leuz and Wysocki (2016) document a substitution (or "chilling") effect of Regulation Fair Disclosure (RegFD) on firms' disclosure behavior.² Our paper complements and contrasts with these studies by examining a different source of changes in capital market participants' information set: an increase in tick size incentivizing investors to engage in fundamental information acquisition. As noted earlier, whether firms increase or decrease earnings guidance when investors engage

more in fundamental information acquisition is an empirical question. In addition to documenting a reduction in earnings guidance after an increase in tick size, results of our channel tests generally indicate that managers indeed reduce earnings guidance in response to investors becoming more informed. We also provide evidence that the impact of tick size changes on management earnings guidance extends to another form of voluntary disclosure, firm-initiated press releases.

Second, we extend the earnings-guidance literature by exploring the role of investors' fundamental information acquisition in managers' disclosure decisions. Recent literature documents how macroeconomic conditions affect investors' information needs and thereby firms' earnings guidance (Nagar et al. 2019, Maslar et al. 2021), as well as how regulations such as the International Financial Reporting Standards, universal demand laws, and Uniform Trade Secrets Act affect firms' earnings guidance (Li and Yang 2016, Bourveau et al. 2018, Glaeser 2018). In this study, we explore the TSPP, which increases investors' fundamental information acquisition and firms' financial reporting quality, to examine how earnings guidance changes in response to an increase in tick size. In a concurrent study, Hope and Liu (2022) examine how changes in stock liquidity induced by the TSPP affect management earnings guidance. Our study differs from theirs in that we advance three alternative mechanisms through which changes in tick size might affect earnings guidance, two of which we empirically verify, whereas Hope and Liu (2022) base their argument solely on stock liquidity.³

Third, our study offers evidence about the indirect consequences of the TSPP, which allows the SEC, market participants, and the public to study and assess the effect of larger tick sizes on the liquidity and stock trading of small-cap stocks. Prior papers (e.g., Rindi and Werner 2019, Albuquerque et al. 2020, Chung et al. 2020) mainly examine the effects of tick size increases on capital market consequences, such as liquidity and market depth. For example, Ye et al. (2023) find that the TSPP increases the investment-q sensitivity of treatment firms and document that the subsequent increase in fundamental information acquisition facilitates managerial learning from stock prices. Our finding that management earnings guidance responds to TSPP-induced increases in investor fundamental information acquisition and firm financial reporting quality suggests that any assessment of the impact of widening tick sizes should include corporate disclosure decisions. An extensive body of literature in accounting and finance documents that voluntary corporate disclosure has capital market consequences (e.g., Diamond and Verrecchia 1991, Healy and Palepu 2001, Kothari 2001). In conjunction with our paper, these studies suggest that the effect of increased tick size on capital market characteristics, such as trading volume

and liquidity, can be enhanced or mitigated by earnings guidance in response to a change in tick size.

2. Background and Hypothesis Development

2.1. Institutional Background

On April 5, 2012, President Barack Obama signed the Jumpstart Our Business Startups (JOBS) Act into law to encourage investment in small businesses. Section 106(b) of the JOBS Act required the SEC to design a study and report to Congress on how the 2001 decimalization (i.e., introduction of the \$0.01 tick size in the U.S. equity market) affected the number of initial public offerings and the liquidity and trading of small-cap companies. Specifically, the JOBS Act authorized the SEC to redesign a minimum tick size increment, greater than \$0.01 but less than \$0.10, for small-cap stocks. In 2014, the SEC directed the Financial Industry Regulatory Association and the national securities exchanges to collectively develop an experimental program to widen the quoting and trading tick size increments for small-cap stocks.⁴

The resulting TSPP began on October 3, 2016, and was phased in gradually until October 31, 2016, and expired on September 28, 2018.⁵ Its primary purpose was to investigate whether increasing the minimum tick size for small-cap stocks would enhance market-making and price discovery for these often lightly traded stocks. Eligible stocks included all those with \$3 billion or less in market capitalization, an average daily trading volume of at least 1 million shares, and a volume-weighted average price of at least \$2.00 for each trading day. Participating firms were randomly selected from eligible stocks using a stratified random sampling process. The three test groups of 400 stocks each had different quoting and trading increments. Test group 1 (TG1) was quoted in \$0.05 increments, but continued to trade at \$0.01 increments. Test group 2 (TG2) was quoted in and traded at \$0.05 minimum increments with certain exemptions for midpoint executions, retail, and negotiated trades. Test group 3 (TG3) followed the same rules as TG2, with an additional “trade-at” requirement.⁶ The control group comprised approximately 1,400 stocks that continued to be traded at and quoted in \$0.01 increments.

Perhaps unsurprisingly, the TSPP’s experimental nature attracted academic interest. Subsequent papers have examined how an increase in tick size affects stock market consequences, such as market liquidity and trading volume (e.g., Rindi and Werner 2019, Albuquerque et al. 2020, Chung et al. 2020). More importantly, increasing tick sizes could reduce the level at which algorithmic trading crowds out fundamental investors (Stiglitz 2014, Ahmed et al. 2020, Lee and Watts 2021), likely because it increases algorithmic trading’s execution costs and reduces algorithmic traders’ speed advantage. When the tick size increases, algorithmic traders must increase the

price even more to obtain price priority. As the per-trade transaction cost increases, algorithmic trading become less profitable, and, thus, algorithmic traders become more reluctant to trade on small-cap stocks. In contrast, increased tick size reduces trading costs for fundamental traders via greater market-making activities and reduced risk of undercutting.⁷ Hence, fundamental investors gain incentives to incur the costs of information acquisition and processing (see Blankespoor et al. 2020). These investors’ trading actions can increase the extent to which stock prices reflect the fundamentals of the underlying firm (Stiglitz 2014). In contrast, algorithmic trading can reduce the information content of stock prices (Weller 2018). Consistent with this argument, recent evidence indicates that the TSPP led to greater fundamental information acquisition by investors (Ahmed et al. 2020, Chung et al. 2020, Lee and Watts 2021, Ye et al. 2023).

2.2. Hypothesis Development

A key objective of volunteering information to investors is to reduce information asymmetry between investors and firms. When trading conditions in the secondary markets (e.g., tick sizes) provide incentives for fundamental investors to acquire information and enhance monitoring, thereby reducing this information asymmetry, firms may be able to avoid costly earnings guidance.

In developing our hypothesis, we assume that investors incentivized by an increase in tick size to acquire information are more inclined to collect information concerning upcoming earnings announcements and future earnings of the firm. This assumption is supported by Lee and Watts (2021), who find that treatment firms’ preannouncement returns better predict standardized unexpected earnings in subsequent quarters and that these firms experience increased EDGAR web traffic prior to earnings announcements. The finding of improved financial reporting quality in Ahmed et al. (2020) similarly suggests that investors scrutinize firms’ earnings. We further assume that an increase in fundamental information acquisition and closer scrutiny of firms’ earnings enable investors to better predict future earnings due to the close link between investors’ ability to predict future earnings and managers’ provision of earnings guidance. According to the traditional investor expectation alignment hypotheses, managers provide earnings guidance to align investors’ earnings expectations with their own, which is based on their private information (Patell 1976, Kasznik and Lev 1995). Hence, one might expect less need for managers to provide earnings guidance when investors can better predict earnings on their own. Earnings guidance has two important roles: (i) to inform investors’ earnings expectations and (ii) to provide additional input into investors’ analyses of the firm. Tension thus arises from the perspective of a firm considering whether to provide earnings guidance.

We begin our discussion of the link between tick size increases and earnings guidance by proposing three reasons for reduced earnings guidance. First, increased tick size motivates fundamental investors to search for and process information more extensively (Lee and Watts 2021). As a result, fundamental investors are likely to develop more accurate expectations about earnings performance, and their trading activities are more likely to result in stock prices that better reflect firm fundamentals. With investors' increased acquisition of fundamental information leading to better alignment of beliefs between managers and outsiders in terms of firm performance, the benefits (e.g., improved stock liquidity and reduced cost of capital) of issuing earnings guidance to reduce information asymmetry between the firm and its investors are likely to diminish (Healy and Palepu 2001, Verrecchia 2001, Beyer et al. 2010, Huang et al. 2021). However, the costs of issuing earnings guidance still remain. For example, issuing earnings guidance may disclose a firm's proprietary information to competitors, which hurts its competitiveness in the product market. Prior studies document that proprietary costs are important disclosure costs that discourage managers from issuing earnings guidance (Healy and Palepu 2001, Verrecchia 2001, Li et al. 2018). Therefore, from a cost-benefit trade-off perspective, managers have less incentive to engage in the potentially costly provision of earnings guidance to inform already informed investors.⁸

Second, Ahmed et al. (2020) find that increased fundamental information acquisition due to the TSPP increases financial reporting quality, consistent with increased scrutiny of managers' financial reporting choices reducing managerial incentives to engage in misreporting. Given the close relationship between mandatory and voluntary disclosures, any changes in earnings guidance may be a response to changes in mandatory disclosure. In particular, the net benefits of a given voluntary disclosure are conditional on the firm's other disclosures and likely depend on similarities between the content of different forms of disclosures (Noh et al. 2019). If investors can better forecast future earnings due to improved mandatory disclosure (e.g., increased financial reporting quality), then managers have less incentive to issue potentially costly earnings guidance to align investors' expectations with their beliefs about future performance.

Third, related to the argument by Ahmed et al. (2020) that fundamental information acquisition increases scrutiny of managers' financial reporting choices, one can also argue that such acquisition increases scrutiny of voluntary disclosure and constrains managers' issuance of misleading earnings guidance. Prior literature documents extensive evidence of the opportunistic use of earnings guidance. For example, managers likely use earnings guidance to manage expectations or mislead investors by lowering investor expectations to beatable levels (Matsumoto 2002, Cotter et al. 2006, Johnson et al.

2020). To the extent that increased scrutiny of voluntary disclosure deters misleading earnings guidance, one might expect more fundamental information acquisition to deter managers from releasing misleading earnings guidance.

Having discussed possible reasons that increased tick size might reduce firms' issuance of earnings guidance, we now discuss possible reasons to expect the opposite result. First, the guidance a firm provides can be regarded as additional input into investors' analyses of the firm. The mosaic theory of securities analysis refers to a common method of collecting various pieces of public and private information, including earnings guidance, to evaluate or monitor a firm (Rogers and Stocken 2005, Kothari et al. 2009, Hutton et al. 2012, Rogers and Van Buskirk 2013). Therefore, when investors want to acquire and analyze a firm's information, the firm might cater to this information demand by issuing more earnings guidance (Kim et al. 2018). Second, managers might issue earnings guidance strategically to obtain equity market feedback, particularly from informed investors, that is useful to their corporate decision-making (Bond et al. 2012, Gao and Liang 2013, Jayaraman and Wu 2020, Chen et al. 2021). To the extent that fundamental information acquisition and information processing increase to make stock prices more efficient, managers might be more inclined to issue earnings guidance after initiation of the TSPP.⁹

In sum, from the perspective of earnings guidance as an expectation-management tool (to either lead or mislead investors), we predict that an increase in tick size will reduce the issuance of earnings guidance if already-informed investors do not need the earnings guidance or are scrutinizing firms' disclosure choices more intensely. In contrast, from the perspective of earnings guidance as additional input provided to investors interested in learning about the firm, we predict that fundamental information acquisition will encourage firms to issue earnings guidance to cater to this information demand. Given the tension in the hypothesis, we formulate our primary hypothesis in a null form as follows:

Hypothesis 1. *An increase in tick size does not affect the issuance of earnings guidance by firms.*¹⁰

3. Sample

3.1. Sample Selection

We obtain the list of stocks included in the TSPP from the website of the Financial Industry Regulatory Association.¹¹ The SEC formally launched the two-year TSPP on October 3, 2016. Therefore, our sample period covers the seven quarters from January 1, 2015, to September 30, 2018, excluding the fourth quarter of 2016. Specifically, we define the period from January 1, 2015, to September 31, 2016, as the prepilot period and the period from January 1, 2017, to September 30, 2018, as the postpilot period.

Prior studies (e.g., Chung et al. 2020, Lee and Watts 2021) find that none of the results related to investors' incentives for fundamental information acquisition varies across the three test groups (i.e., TG1, TG2, and TG3) in an economically meaningful way. Therefore, to increase the power of our tests, we combine the three test groups and define them jointly as the treatment group. We collect data on earnings guidance from I/B/E/S, firm financial information from Compustat, and stock information from CRSP. Analyst coverage and institutional ownership data are from the I/B/E/S and Thomson Reuters Institutional Holdings (13f) databases, respectively.

Following prior studies (e.g., Rindi and Werner 2019, Ahmed et al. 2020, Lee and Watts 2021), we exclude stocks that enter the TSPP after October 31, 2016; change their test groups during the TSPP; are dropped from the TSPP due to a merger, symbol change, delisting, or price drop below \$1; or cannot be matched in Compustat. Because firms in the utility and finance industries (Standard Industrial Classification 4400–5000 and 6000–6999) may have different incentives for issuing earnings guidance, we also exclude these firms from the analysis. We further exclude firms with insufficient data to calculate the control variables or with only one observation in our sample. To mitigate the effect of outliers, we winsorize all continuous variables at the 1st and 99th percentiles. Our final sample thus consists of 14,157 firm-quarter observations for 1,107 unique firms. Of these, 7,048 observations are in the treatment group, and 7,109 observations are in the control group. Table 1 presents the distribution of the observations, which shows that the observations are almost equally distributed between the two groups in each quarter of our sample period.

3.2. Research Design

Following Ahmed et al. (2020) and Lee and Watts (2021), we use a difference-in-differences design to investigate

our research question. The regression specification is as follows:

$$\begin{aligned} DumMF_{i,t}/FreqMF_{i,t} = & \beta_1 + \beta_2 Treat_i \times Post_t \\ & + \sum_{k=1}^K \gamma_k Controls_{i,t-1} + Firm\ FE \\ & + Year - Quarter\ FE + \varepsilon_{i,t}, \quad (1) \end{aligned}$$

where i denotes the firm, t the year, and ε the error term. The dependent variables are two management guidance variables: forecast likelihood ($DumMF$) and forecast frequency ($FreqMF$). We include both annual and quarterly earnings guidance.¹² Forecast likelihood ($DumMF$) is a dummy variable that equals one if the firm issues at least one earnings guidance during the current quarter, and zero otherwise. Forecast frequency ($FreqMF$) is the natural logarithm of one plus the number of earnings guidance the firm issues in the current quarter. The postpilot dummy ($Post$) takes the value of one for the postpilot period (i.e., January 1, 2017, to September 30, 2018), and zero for the prepilot period (i.e., January 1, 2015, to September 31, 2016). The treatment dummy ($Treat$) takes the value of one if a firm belongs to the treatment group, and zero if it belongs to the control group. The coefficient on $Treat \times Post$ captures the change in earnings guidance for treatment firms relative to control firms after initiation of the TSPP. We use an ordinary least squares (OLS) model for both the forecast likelihood and the forecast frequency.¹³ We also adjust standard errors for heteroscedasticity and clustering at the firm level.

Following the literature (e.g., Bourveau et al. 2018, Kim et al. 2018), we include several variables as controls, including firm size ($Size$), book-to-market (BM), return on assets (ROA), operating loss ($Loss$), leverage (Lev), institutional ownership (IO), analyst following ($AnalystCov$), stock return volatility ($StkVol$), earnings volatility ($EarnVol$), stock return (Ret), stock turnover ($Turnover$),

Table 1. Sample Distribution

Year-quarter	Obs. of treatment firms	Obs. of control firms	Total obs.
2015Q1	462	461	923
2015Q2	480	474	954
2015Q3	483	483	966
2015Q4	520	522	1,042
2016Q1	500	509	1,009
2016Q2	509	516	1,025
2016Q3	512	487	999
2017Q1	514	544	1,058
2017Q2	513	514	1,027
2017Q3	512	512	1,024
2017Q4	513	511	1,024
2018Q1	531	544	1,075
2018Q2	530	547	1,077
2018Q3	470	485	955
Total	7,048	7,109	14,157

Notes. This table presents the year-quarter distribution of observations by treatment and control firms. The sample period is from January 1, 2015, to September 30, 2018, excluding the fourth quarter of 2016. The appendix presents the variable definitions.

cash holdings (*Cash*), asset tangibility (*Tangibility*), equity issuance (*EquityIssue*), debt issuance (*DebtIssue*), and litigation risk (*Litigation*). Next, following recent studies on the outcomes of the TSPP (e.g., Chung et al. 2020, Lee and Watts 2021), we control for contemporaneous values of stock price synchronicity (*Sync*) and stock liquidity (*Liquidity*).¹⁴ See the appendix for detailed variable definitions. We also include firm fixed effects (*Firm FE*) and year-quarter fixed effects (*Year-Quarter FE*) to control for unobserved heterogeneity across firms and over time.

3.3. Descriptive Statistics

Table 2, Panel A presents the summary statistics for the variables in the baseline analysis. The mean value of *DumMF* is 0.250, indicating that 25% of firm-quarter observations have at least one earnings guidance during our sample period. *FreqMF* has a mean value of 0.211, which indicates that the average frequency of earnings guidance is 0.236 times per quarter. As all firms in the TSPP are small-cap, the mean value of *Size* is 5.861, indicating mean assets of 885 million U.S. dollars. The mean (median) value of *ROA* is -0.016 (0.005), and about 41.2% of firm-quarters in our sample have an operating loss. The summary statistics for the control variables are largely consistent with those of prior studies (e.g., Ahmed et al. 2020, Lee and Watts 2021).

Table 2, Panel B presents the correlations between the variables in the baseline analysis. *Size* and *ROA* both positively correlate with the two earnings guidance variables (i.e., *DumMF* and *FreqMF*), indicating that larger firms and those with a better performance are more likely to issue earnings guidance. *IO* and *AnalystCov* also positively correlate with the two earnings guidance variables, implying that firms with higher percentages of institutional ownership and more analyst coverage tend to issue more earnings guidance. Lastly, *StkVol* and *EarnVol* negatively correlate with the two earnings guidance variables, suggesting that firms with higher uncertainty in operations are less likely to issue earnings guidance.

4. Empirical Analysis of the Effect of the TSPP on Earnings Guidance

4.1. Baseline Analysis

We begin by plotting the patterns of earnings guidance separately for the treatment and control firms. Specifically, we compute the mean values of *DumMF* and *FreqMF* for the treatment and control groups in each quarter during our sample period and plot the mean values in separate graphs in Figure 1. The graphs show a decreasing trend in both *DumMF* and *FreqMF* for the treatment firms. No such trend is observed for the control firms. The findings provide preliminary evidence that after TSPP implementation, earnings guidance decreases for treatment firms, but not for control firms.

Next, we perform a regression analysis on the effect of the TSPP on earnings guidance using the specification in Equation (1). We report the regression results in Table 3. We use *DumMF* as the dependent variable for the results in columns (1) and (2) and *FreqMF* as the dependent variable for the results in columns (3) and (4). Columns (1) and (3) report the results without firm and year-quarter fixed effects, and columns (2) and (4) report the results of the baseline model in Equation (1) that include these two fixed effects. In all four columns, the coefficients on *Treat* $\times$ *Post* are negative and significant at the 5% level. The results suggest a significant decrease in earnings guidance for treatment firms relative to control firms after TSPP initiation. In terms of economic significance, the forecast likelihood diminishes by 0.027 for treatment firms relative to control firms. Given that the mean value of this variable is 0.25 in our sample, this constitutes a 10.8% drop relative to the mean. The corresponding percentage drop for the forecast frequency is 10.4%.¹⁵ The coefficients on the control variables are generally consistent with the literature. Collectively, the results in the baseline analysis document a significant decrease in earnings guidance for treatment firms relative to control firms after TSPP implementation.

4.2. Parallel-Trend Test

The key assumption of our difference-in-differences model is that in the absence of increased tick size from the TSPP, the average change in earnings guidance should be the same for both the treatment and control groups. To validate this assumption, we follow Bourveau et al. (2018) and Ahmed et al. (2020) and adopt a parallel-trend analysis. This analysis allows us to investigate trends in earnings guidance prior to the TSPP. Specifically, we decompose our sample period into separate quarters and generate a dummy variable for each quarter. We interact each of the quarter dummies with *Treat* and replace *Treat* $\times$ *Post* with these interaction terms in the specification in Equation (1).

The results are reported in Table 4. In both columns, the coefficients on *Treat* $\times$ *Pre6*, *Treat* $\times$ *Pre5*, *Treat* $\times$ *Pre4*, *Treat* $\times$ *Pre3*, *Treat* $\times$ *Pre2*, and *Treat* $\times$ *Pre1* are not significant, implying that, generally, no material difference in earnings guidance exists between treatment and control firms before initiation of the TSPP. Further, we find that the effect of TSPP initiation in the quarter immediately after is insignificant, which indicates that the decrease in earnings guidance does not start immediately after initiation of the TSPP. The differences in earnings guidance in the later quarters are all significant and negative, however, suggesting that treatment firms start reducing their earnings guidance in the second quarter after the TSPP's initiation.¹⁶ In Figure 2, we plot the regression coefficients to better visualize the effects of the TSPP on earnings guidance over time. Overall, the findings suggest

Table 2. Descriptive Statistics

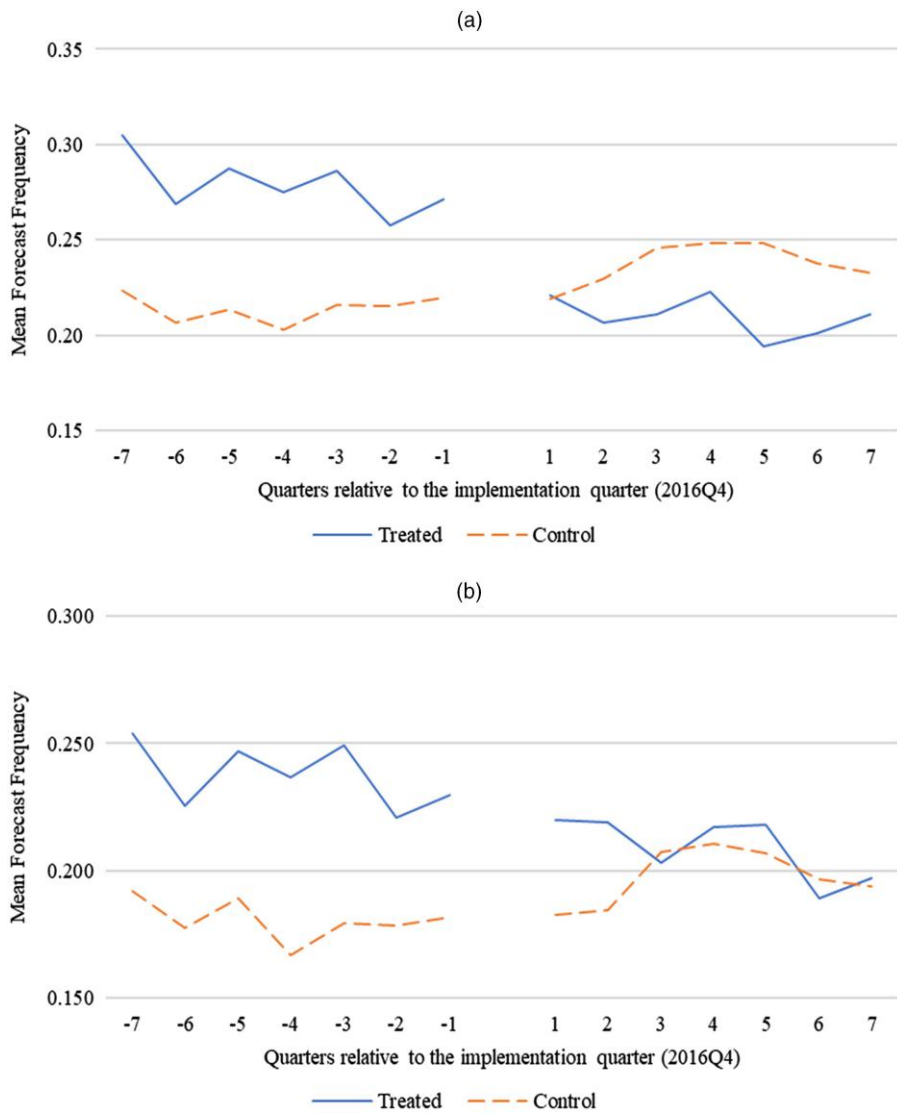
Panel A. Summary statistics																				
Variable	Mean	S.D.	P25	Median	P75															
DumMF _t	0.250	0.433	0.000	0.000	0.000															
FreqMF _t	0.211	0.382	0.000	0.000	0.000															
Size _{t-1}	5.861	1.481	4.806	5.931	6.989															
BM _{t-1}	0.489	0.468	0.220	0.404	0.660															
ROA _{t-1}	-0.016	0.066	-0.020	0.005	0.016															
Loss _{t-1}	0.412	0.492	0.000	0.000	1.000															
Lev _{t-1}	0.185	0.216	0.000	0.116	0.303															
IO _{t-1}	0.625	0.293	0.397	0.692	0.879															
AnalystCov _{t-1}	1.392	0.787	0.693	1.609	1.946															
StkVol _{t-1}	0.059	0.034	0.037	0.051	0.073															
EarnVol _{t-1}	0.027	0.048	0.006	0.012	0.027															
Ret _{t-1}	0.007	0.057	-0.024	0.004	0.033															
Turnover _{t-1}	0.464	0.447	0.191	0.349	0.582															
Cash _{t-1}	0.260	0.277	0.041	0.152	0.397															
Tangbilty _{t-1}	0.207	0.226	0.046	0.119	0.282															
EquityIssue _{t-1}	0.472	0.499	0.000	0.000	1.000															
DebtIssue _{t-1}	0.306	0.461	0.000	0.000	1.000															
Litigation _{t-1}	0.089	0.125	0.001	0.018	0.142															
Sync _t	-0.261	1.002	-0.909	-0.232	0.420															
Illiquidity _t	1.388	10.980	0.002	0.007	0.050															

Panel B. Correlation matrix																				
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
(1) DumMF	1.000																			
(2) FreqMF	0.957*	1.000																		
(3) Size	0.284*	0.266*	1.000																	
(4) BM	-0.056*	-0.065*	0.078*	1.000																
(5) ROA	0.186*	0.173*	0.367*	0.123*	1.000															
(6) Loss	-0.180*	-0.162*	-0.278*	-0.013	-0.626*	1.000														
(7) Lev	-0.011	-0.018*	0.405*	-0.156*	0.071*	-0.057*	1.000													
(8) IO	0.354*	0.345*	0.572*	-0.140*	0.179*	-0.132*	0.116*	1.000												
(9) AnalystCov	0.307*	0.307*	0.475*	-0.204*	-0.029*	0.039*	0.160*	0.590*	1.000											
(10) StkVol	-0.199*	-0.195*	-0.275*	-0.020*	-0.419*	0.364*	-0.030*	-0.168*	-0.001	1.000										
(11) EarnVol	-0.155*	-0.148*	-0.302*	-0.148*	-0.419*	0.286*	-0.097*	-0.177*	-0.068*	0.316*	1.000									
(12) Ret	-0.011	-0.008	-0.027*	-0.036*	-0.046*	0.040*	0.007	0.017*	0.002	0.083*	0.028*	1.000								
(13) Turnover	0.050*	0.048*	0.095*	-0.115*	-0.197*	0.128*	0.009	0.256*	0.335*	0.405*	0.173*	0.023*	1.000							
(14) Cash	-0.166*	-0.147*	-0.443*	-0.231*	-0.532*	0.391*	-0.351*	-0.084*	0.055*	0.357*	0.369*	0.043*	0.215*	1.000						
(15) Tangibility	-0.084*	-0.088*	0.244*	0.254*	0.181*	-0.119*	0.290*	-0.068*	-0.038*	-0.103*	-0.170*	-0.027*	-0.079*	-0.425*	1.000					
(16) EquityIssue	-0.090*	-0.081*	-0.199*	-0.140*	-0.283*	0.222*	-0.122*	-0.038*	0.117*	0.201*	0.181*	0.019*	0.118*	0.346*	-0.131*	1.000				
(17) DebtIssue	0.076*	0.066*	0.225*	0.054*	0.074*	-0.081*	0.270*	0.058*	0.049*	-0.088*	-0.087*	-0.032*	-0.026*	-0.313*	0.162*	-0.108*	1.000			
(18) Litigation	0.007	0.002	0.163*	0.011	-0.045*	0.056*	0.089*	0.115*	0.142*	0.180*	0.023*	0.004	0.155*	0.005	0.000	0.026*	0.031*	1.000		
(19) Sync	0.098*	0.094*	0.213*	-0.008	0.058*	-0.065*	0.031*	0.184*	0.152*	-0.043*	-0.049*	-0.027*	0.027*	-0.044*	-0.009	-0.032*	0.027*	0.032*	1.000	
(20) Illiquidity	-0.050*	-0.049*	-0.173*	0.123*	0.007	0.025*	-0.041*	-0.21*	-0.195*	-0.001	-0.010	0.015*	-0.104*	-0.036*	0.001	-0.053*	-0.032*	-0.061*	-0.050*	1.000

Notes. This table presents the summary statistics and correlation matrix of the variables in the baseline analysis. The sample period is from January 1, 2015, to September 30, 2018, excluding the fourth quarter of 2016. The appendix presents the variable definitions.

* $p < 0.10$.

Figure 1. (Color online) Patterns of Earnings Guidance around the Implementation of the Tick Size Pilot Program



Notes. This figure plots the mean values of the likelihood (panel (a)) and frequency (panel (b)) of management earnings guidance separately for the treatment firms and the control firms. The sample period is from January 1, 2015, to September 30, 2018, excluding the fourth quarter of 2016. The appendix presents the variable definitions. (a) Likelihood of earnings guidance. (b) Frequency of earnings guidance.

that the parallel-trend assumption behind the difference-in-differences model holds in our setting.

4.3. Robustness Tests

In addition to the parallel-trend test, we perform tests to further examine the robustness of our findings and present the results in Table 5. First, we use a sample restricted to only firms with at least one earnings guidance over our sample period and present the results in Table 5, Panel A. This approach helps alleviate the concern that our findings are due to the fact that small firms are less likely than large firms to issue guidance.¹⁷ Despite a significant reduction in sample size, we continue to find negative and significant coefficients on *Treat* × *Post*.¹⁸ Second, we examine whether our findings hold

using different event windows and document the results in Table 5, Panel B. We use three alternative event windows that include the four, six, and eight quarters before and after the program, and the results remain consistent.

In the third test, we examine the sensitivity of our findings to a propensity-score-matched (PSM) sample. In forming the PSM sample, we regress *Treat* on the covariates in our baseline model using data in the quarter immediately prior to the implementation of the program (i.e., *t* − 1) and compute the propensity score based on the regression. Next, we match (without replacement) each firm in the treatment group to one firm in the control group with the closest propensity score. We then re-estimate our baseline model in Equation (1) based on the PSM sample and document the results in Table 5,

Table 3. Tick Size Pilot Program and Earnings Guidance

Variable	DumMF _{<i>t</i>}		FreqMF _{<i>t</i>}	
	(1)	(2)	(3)	(4)
<i>Treat</i> × <i>Post</i>	−0.034** (−2.577)	−0.027** (−2.238)	−0.027** (−2.241)	−0.022** (−2.004)
<i>Treat</i>	0.044** (1.968)		0.041** (2.033)	
<i>Post</i>	0.018* (1.848)		0.012 (1.304)	
<i>Size</i> _{<i>t</i>−1}	0.012 (1.214)	0.017 (1.561)	0.008 (0.906)	0.017* (1.697)
<i>BM</i> _{<i>t</i>−1}	−0.029 (−1.641)	−0.022* (−1.904)	−0.031** (−2.076)	−0.022** (−2.074)
<i>ROA</i> _{<i>t</i>−1}	−0.018 (−0.153)	0.123*** (2.675)	0.010 (0.097)	0.104*** (2.623)
<i>Loss</i> _{<i>t</i>−1}	−0.060*** (−3.429)	−0.007 (−1.279)	−0.041*** (−2.613)	−0.007 (−1.256)
<i>Lev</i> _{<i>t</i>−1}	−0.255*** (−5.043)	0.023 (0.672)	−0.224*** (−5.335)	0.035 (1.077)
<i>IO</i> _{<i>t</i>−1}	0.244*** (5.602)	0.045 (1.114)	0.209*** (5.714)	0.048 (1.247)
<i>AnalystCov</i> _{<i>t</i>−1}	0.118*** (7.472)	0.028*** (2.697)	0.108*** (7.710)	0.025*** (2.528)
<i>StkVol</i> _{<i>t</i>−1}	−0.957*** (−6.008)	−0.028 (−0.421)	−0.907*** (−6.317)	−0.052 (−0.827)
<i>EarnVol</i> _{<i>t</i>−1}	−0.206** (−2.004)	−0.033 (−0.802)	−0.202** (−2.142)	−0.051 (−1.284)
<i>Ret</i> _{<i>t</i>−1}	0.009 (0.198)	−0.017 (−0.677)	0.027 (0.709)	−0.010 (−0.430)
<i>Turnover</i> _{<i>t</i>−1}	0.008 (0.482)	0.003 (0.321)	0.004 (0.292)	0.000 (0.042)
<i>Cash</i> _{<i>t</i>−1}	−0.258*** (−6.337)	−0.011 (−0.241)	−0.210*** (−5.844)	−0.020 (−0.487)
<i>Tangibility</i> _{<i>t</i>−1}	−0.250*** (−4.923)	−0.046 (−0.584)	−0.210*** (−4.851)	−0.040 (−0.576)
<i>EquityIssue</i> _{<i>t</i>−1}	−0.034** (−2.082)	−0.006 (−0.760)	−0.030** (−2.098)	−0.003 (−0.511)
<i>DebtIssue</i> _{<i>t</i>−1}	0.033** (2.391)	−0.001 (−0.112)	0.026** (2.176)	−0.001 (−0.178)
<i>Litigation</i> _{<i>t</i>−1}	−0.068** (−2.126)	−0.005 (−0.361)	−0.066** (−2.441)	−0.006 (−0.454)
<i>Sync</i> _{<i>t</i>}	0.005 (1.364)	0.002 (1.151)	0.005 (1.284)	0.002 (1.242)
<i>Illiquidity</i> _{<i>t</i>}	0.001* (1.725)	0.000 (1.565)	0.001* (1.875)	0.000 (1.524)
Firm FE	No	Yes	No	Yes
Year-quarter FE	No	Yes	No	Yes
<i>N</i>	14,157	14,157	14,157	14,157
Adjusted <i>R</i> ²	0.215	0.830	0.205	0.800

Notes. This table presents the baseline results of the effect of the Tick Size Pilot Program on the likelihood and frequency of earnings guidance. The regressions are performed using OLS. The *t*-statistics in parentheses are adjusted for heteroscedasticity and clustered by firm. The appendix presents the variable definitions.

p* < 0.10; *p* < 0.05; ****p* < 0.01.

Panel C. The negative impact of the TSPP on earnings guidance persists in the PSM sample, suggesting that our results are unlikely driven by sample selection bias.

For the final robustness check, we conduct two placebo tests to examine whether our findings in the baseline analysis are obtained simply by chance. In the first

Table 4. Parallel-Trend Test

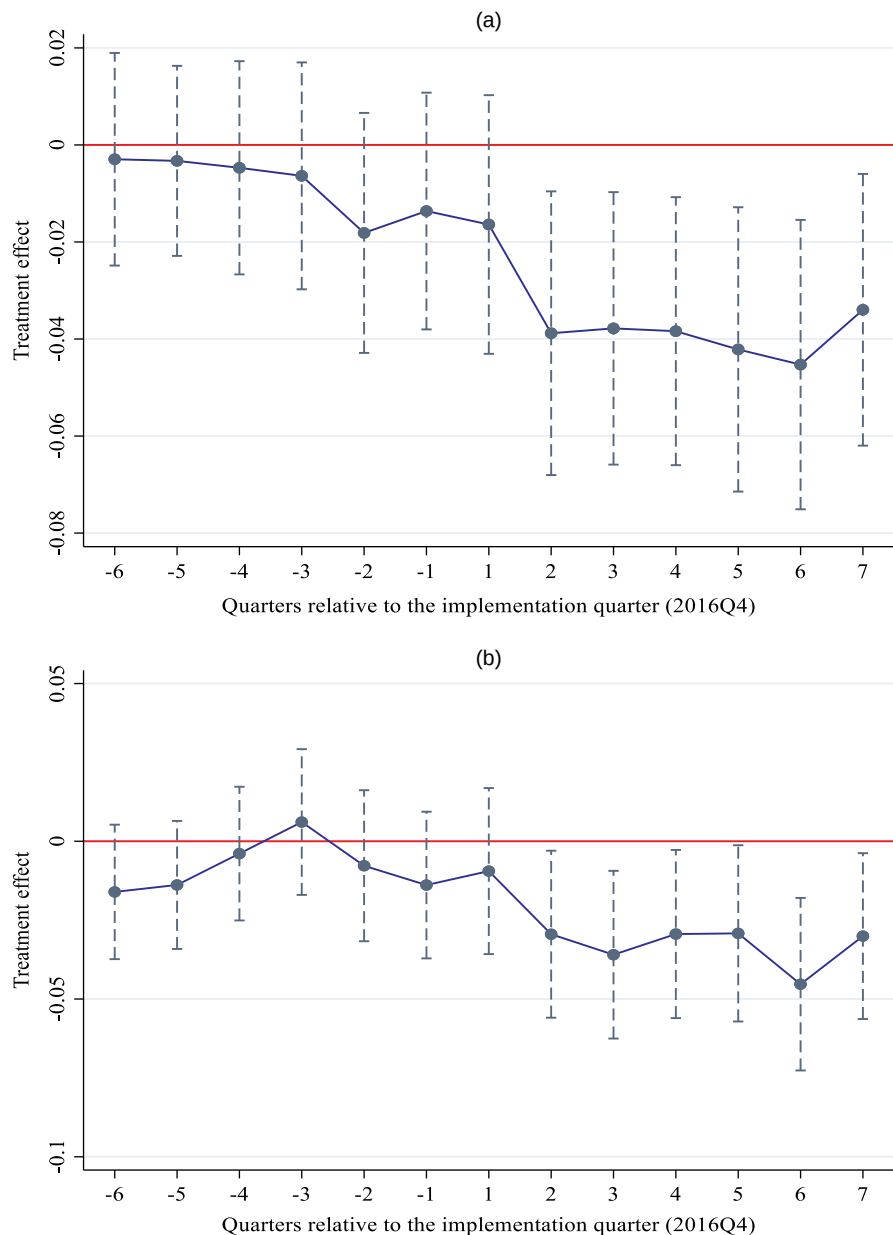
Variable	DumMF _{<i>t</i>}	FreqMF _{<i>t</i>}
	(1)	(2)
<i>Treat</i> × <i>Pre</i> ₆	−0.003 (−0.245)	−0.013 (−1.070)
<i>Treat</i> × <i>Pre</i> ₅	−0.004 (−0.345)	−0.011 (−0.903)
<i>Treat</i> × <i>Pre</i> ₄	−0.005 (−0.397)	−0.003 (−0.255)
<i>Treat</i> × <i>Pre</i> ₃	−0.007 (−0.513)	0.008 (0.582)
<i>Treat</i> × <i>Pre</i> ₂	−0.019 (−1.238)	−0.006 (−0.430)
<i>Treat</i> × <i>Pre</i> ₁	−0.014 (−0.966)	−0.013 (−0.935)
<i>Treat</i> × <i>Post</i> ₁	−0.017 (−1.043)	−0.009 (−0.561)
<i>Treat</i> × <i>Post</i> ₂	−0.039** (−2.197)	−0.028* (−1.779)
<i>Treat</i> × <i>Post</i> ₃	−0.038** (−2.228)	−0.035** (−2.215)
<i>Treat</i> × <i>Post</i> ₄	−0.038** (−2.281)	−0.028* (−1.738)
<i>Treat</i> × <i>Post</i> ₅	−0.042** (−2.350)	−0.027 (−1.639)
<i>Treat</i> × <i>Post</i> ₆	−0.045** (−2.474)	−0.042*** (−2.598)
<i>Treat</i> × <i>Post</i> ₇	−0.033* (−1.945)	−0.027* (−1.742)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year-quarter FE	Yes	Yes
<i>N</i>	14,157	14,157
Adjusted <i>R</i> ²	0.830	0.800

Notes. This table presents the results of the parallel-trend test. The regressions are performed using OLS. The *t*-statistics in parentheses are adjusted for heteroscedasticity and clustered by firm. The appendix presents the variable definitions.

p* < 0.10; *p* < 0.05; ****p* < 0.01.

placebo test, we follow Ahmed et al. (2020) and pick April 1, 2015, as the pseudo-event date, while maintaining the same treatment and control groups as in our baseline analysis. Accordingly, we define the period from July 1, 2013, to March 31, 2015, as the pseudo-prepilot period and the period from April 1, 2015, to December 31, 2016, as the pseudo-postpilot period. *Pseudo_Post* is a dummy variable that equals one for the pseudo-postpilot period, and zero otherwise. We re-estimate our baseline regression in Equation (1) by replacing *Post* with *Pseudo_Post*. Table 5, Panel D reports the results of the placebo tests. The coefficient on *Treat* × *Pseudo_Post* is not statistically different from zero, indicating no significant change in earnings guidance for treatment firms around the artificially chosen data. In the second placebo test, we randomly assign our sample firms into the treatment and control groups and generate *Pseudo_Treat* as an indicator for firms in the pseudo-treatment group. We then re-estimate the baseline model by replacing *Treat* × *Post* with *Pseudo_Treat* × *Post*. We repeat this process 1,000 times, generating 1,000 different coefficient estimates.

Figure 2. (Color online) Treatment Effect of the Tick Size Pilot Program on Earnings Guidance



Notes. This figure plots the treatment effect of the Tick Size Pilot Program on the likelihood (panel (a)) and frequency (panel (b)) of earnings guidance. The sample period is from January 1, 2015, to September 30, 2018, excluding the fourth quarter of 2016. The benchmark quarter is the first quarter of 2014 (i.e., quarter $t - 7$). The dashed line plots the 95% confidence intervals of the treatment effect. The appendix presents the variable definitions. (a) Likelihood of earnings guidance. (b) Frequency of earnings guidance.

Table 5, Panel E documents the distribution of coefficients. Importantly, we find that the actual coefficients on $Treat \times Post$ in Table 3 lie far beyond the lower tail of the distributions of $Pseudo_Treat \times Post$. Overall, the results of the placebo tests further validate our findings in the baseline analysis by confirming that they are not obtained by chance.

5. Potential Channels

In developing our hypothesis, we propose three mechanisms by which the TSPP could lead to a reduction in

earnings guidance: (1) an increase in fundamental information acquisition that makes investors more informed; (2) a substitution between mandatory disclosure and earnings guidance; and (3) managers' reduced motif of issuing earnings guidance to mislead investors. In this section, we test each of these channels and report the results of the channel tests in Table 6.

5.1. Fundamental Information Acquisition

In our first channel test, we examine the role of the increase in fundamental information acquisition on the

Table 5. Robustness Checks

Panel A. Sample using firms with at least one guidance during our sample period						
Variable	$DumMF_t$ (1)		$FreqMF_t$ (2)			
$Treat \times Post$	−0.078** (−2.221)		−0.064* (−1.963)			
Controls	Yes		Yes			
Firm FE	Yes		Yes			
Year-quarter FE	Yes		Yes			
N	4,649		4,649			
Adjusted R^2	0.487		0.486			
Panel B. Alternative event windows						
Variable	Window period					
	[+4, −4]		[+6, −6]		[+8, −8]	
	$DumMF_t$ (1)	$FreqMF_t$ (2)	$DumMF_t$ (3)	$FreqMF_t$ (4)	$DumMF_t$ (5)	$FreqMF_t$ (6)
$Treat \times Post$	−0.021* (−1.803)	−0.022** (−1.971)	−0.024** (−1.991)	−0.020* (−1.725)	−0.028*** (−2.612)	−0.023** (−2.252)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
N	8,144	8,144	12,226	12,226	16,077	16,077
Adjusted R^2	0.843	0.806	0.830	0.795	0.829	0.793
Panel C. Propensity-score matching						
Variable	$DumMF_t$ (1)		$FreqMF_t$ (2)			
$Treat \times Post$	−0.035*** (−2.700)		−0.026** (−2.097)			
Controls	Yes		Yes			
Firm FE	Yes		Yes			
Year-quarter FE	Yes		Yes			
N	10,885		10,885			
Adjusted R^2	0.836		0.809			
Panel D. Placebo tests: Pseudo-event date						
Variable	$DumMF_t$ (1)		$FreqMF_t$ (2)			
$Treat \times Pseudo_Post$	0.006 (0.573)		0.002 (0.195)			
Controls	Yes		Yes			
Firm FE	Yes		Yes			
Year-Quarter FE	Yes		Yes			
N	12,658		12,658			
Adjusted R^2	0.851		0.820			
Panel E. Placebo tests: Pseudo-treatment firms						
Variable	$DumMF_t$ Coefficient on $Pseudo_Treat \times Post$ (1)		$FreqMF_t$ Coefficient on $Pseudo_Treat \times Post$ (2)			
Minimum	−0.010		−0.010			
10%	−0.004		−0.004			
20%	−0.002		−0.002			
30%	−0.001		−0.001			
40%	0.000		0.000			
Median	0.001		0.001			
60%	0.003		0.003			
70%	0.004		0.004			

Table 5. (Continued)

Panel E. Placebo tests: Pseudo-treatment firms		
Variable	$DumMF_t$ Coefficient on $Pseudo_Treat \times Post$ (1)	$FreqMF_t$ Coefficient on $Pseudo_Treat \times Post$ (2)
80%	0.005	0.005
90%	0.007	0.006
Maximum	0.014	0.014
Mean	0.001	0.001
Actual coefficient on $Treat \times Post$ in columns (2) and (4) of Table 3	−0.027	−0.022

Notes. This table presents the results of the robustness checks. The regressions are performed using OLS. The t -statistics in parentheses are adjusted for heteroscedasticity and clustered by firm. The appendix presents the variable definitions.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

relation between the TSPP and earnings guidance. Following prior studies (e.g., Ahmed et al. 2020, Lee and Watts 2021, Ye et al. 2023), we measure fundamental information acquisition using nonrobotic downloads of firms' SEC EDGAR filings and the following three methods: *D RTPV* from Drake et al. (2015), Loughran and McDonald's (2017) *LMPV*, and Ryans' (2017, 2021) *RPV*. Higher values of these three measures imply greater investor fundamental information acquisition.¹⁹ Following Ye et al. (2023), we first calculate the average value of the three fundamental information-acquisition measures for each firm in the prepilot and postpilot periods. Then, we compute the change in the average values of the three measures during the two periods. *Treat_FIAIncMore* (*Treat_FIAIncLess*) is a dummy variable that equals one for treatment firms that experience a change in average fundamental information acquisition above (below) the sample median value of all treatment firms, and zero otherwise.²⁰ We re-estimate the baseline model in Equation (1) by replacing $Treat \times Post$ with $Treat_FIAIncMore \times Post$ and $Treat_FIAIncLess \times Post$.

We report the results of this test in Table 6, Panel A. For all three fundamental information-acquisition measures, the coefficients on $Treat_FIAIncMore \times Post$ are significantly negative, whereas the coefficients on $Treat_FIAIncLess \times Post$ are not statistically significant. The differences between the coefficient estimates on $Treat_FIAIncMore \times Post$ and $Treat_FIAIncLess \times Post$ are generally statistically significant. In sum, the results support our argument that an increase in fundamental information acquisition resulting from the TSPP reduces investors' need for earnings guidance and, hence, disincentivizes managers from issuing earnings guidance.

5.2. Substitution between Voluntary and Mandatory Disclosures

The second channel we propose for the negative impact of the TSPP on earnings guidance is the potential substitution between voluntary disclosure (i.e., earnings guidance) and mandatory disclosure (i.e., financial reporting quality). Following Ahmed et al. (2020), as a proxy for

financial reporting quality, we use the absolute value of residuals (*AbsM/JDA*) from estimating Jones' (1991) model modified by Dechow et al. (1995) and Kothari et al. (2005). We further adjust the model following Collins et al. (2017) to account for nonlinearities in performance and growth associated with quarterly data. We calculate the change in the average value of financial reporting quality for each firm in the treatment group from the prepilot to the postpilot periods. Then, we define *Treat_FRQualityIncMore* (*Treat_FRQualityIncLess*) as a dummy variable that equals one for treatment firms that experience a change in average financial reporting quality above (below) the sample median of all treatment firms, and zero otherwise. Next, we re-estimate the baseline model in Equation (1) by replacing $Treat \times Post$ with $Treat_FRQualityIncMore \times Post$ and $Treat_FRQualityIncLess \times Post$.

The regression results are reported in Table 6, Panel B. The panel shows that in both columns, the coefficients on $Treat_FRQualityIncMore \times Post$ are significantly negative, whereas the coefficients on $Treat_FRQualityIncLess \times Post$ are not significant. The differences between the coefficient estimates are statistically significant. Overall, our findings support the argument that an increase in financial reporting quality arising from the TSPP reduces managers' incentives to provide earnings guidance, consistent with a substitution between voluntary and mandatory disclosures (Noh et al. 2019).

5.3. Voluntary Disclosure and Expectations Management

In the last channel, we propose that a decline in earnings guidance caused by the TSPP could be due to a decrease in managers' incentives to manage investors' expectations. Following Johnson et al. (2020), we compute a composite measure of expectations management (*EMI*) as the first principal component of four variables: number of analysts providing forecasts about the firm, percentage of institutional investors' holdings, the firm's unsustainable growth rate, and the firm's solvency. Then, we calculate the average value of expectations management for each firm in the treatment group during the prepilot period.

Table 6. Tests of Possible Mechanisms

Panel A. Fundamental information acquisition						
	<i>Treat_FIAIncMore</i> and <i>Treat_FIAIncLess</i> measured by:					
	<i>DRTPV</i>		<i>LMPV</i>		<i>RPV</i>	
Variable	<i>DumMF_t</i> (1)	<i>FreqMF_t</i> (2)	<i>DumMF_t</i> (3)	<i>FreqMF_t</i> (4)	<i>DumMF_t</i> (5)	<i>FreqMF_t</i> (6)
<i>Treat_FIAIncMore</i> × <i>Post</i> [a]	−0.041** (−2.422)	−0.037** (−2.242)	−0.036** (−2.093)	−0.037** (−2.213)	−0.036** (−2.113)	−0.033** (−2.028)
<i>Treat_FIAIncLess</i> × <i>Post</i> [b]	−0.005 (−0.417)	−0.002 (−0.137)	−0.011 (−0.921)	−0.004 (−0.304)	−0.011 (−0.846)	−0.006 (−0.492)
<i>p</i> -value of F-test: [a] = [b]	0.033	0.030	0.151	0.045	0.145	0.093
Controls	Yes	Yes	Yes	Yes	Yes	Yes
<i>Treat_FIAIncMore</i> × Controls	Yes	Yes	Yes	Yes	Yes	Yes
<i>Treat_FIAIncLess</i> × Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	13,913	13,913	13,913	13,913	13,913	13,913
Adjusted <i>R</i> ²	0.830	0.801	0.830	0.801	0.830	0.801
Panel B. Financial reporting quality						
Variable	<i>DumMF_t</i> (1)		<i>FreqMF_t</i> (2)			
<i>Treat_FRQualityIncMore</i> × <i>Post</i> [a]	−0.043*** (−2.827)		−0.042*** (−2.784)			
<i>Treat_FRQualityIncLess</i> × <i>Post</i> [b]	−0.004 (−0.290)		−0.001 (−0.063)			
<i>p</i> -value of F-test: [a] = [b]	0.021		0.011			
Controls	Yes		Yes			
<i>Treat_FRQualityIncMore</i> × Controls	Yes		Yes			
<i>Treat_FRQualityIncLess</i> × Controls	Yes		Yes			
Firm FE	Yes		Yes			
Year-quarter FE	Yes		Yes			
<i>N</i>	13,946		13,946			
Adjusted <i>R</i> ²	0.830		0.800			
Panel C. Expectations management						
Variable	<i>DumMF_t</i> (1)		<i>FreqMF_t</i> (2)			
<i>Treat_HighEMI</i> × <i>Post</i> [a]	−0.012 (−0.618)		−0.014 (−0.718)			
<i>Treat_LowEMI</i> × <i>Post</i> [b]	−0.011 (−0.918)		−0.004 (−0.359)			
<i>p</i> -value of F-test: [a] = [b]	0.966		0.606			
Controls	Yes		Yes			
<i>Treat_HighEMI</i> × Controls	Yes		Yes			
<i>Treat_LowEMI</i> × Controls	Yes		Yes			
Firm FE	Yes		Yes			
Year-quarter FE	Yes		Yes			
<i>N</i>	11,509		11,509			
Adjusted <i>R</i> ²	0.844		0.818			

Notes. This table presents the results of tests on the possible mechanisms through which the Tick Size Pilot Program affects earnings guidance. The regressions are performed using OLS. The *t*-statistics in parentheses are adjusted for heteroscedasticity and clustered by firm. The appendix presents the variable definitions.

p* < 0.05; *p* < 0.01.

We define *Treat_HighEMI* (*Treat_LowEMI*) as a dummy variable that equals one for treatment firms with average expectations management above (below) the sample median of all treatment firms, and zero otherwise. Once

again, we re-estimate the baseline model in Equation (1) by replacing *Treat* × *Post* with *Treat_HighEMI* × *Post* and *Treat_LowEMI* × *Post* and report the results in Table 6, Panel C. In both columns, the coefficients on

$Treat_HighEMI \times Post$ and $Treat_LowEMI \times Post$ are not statistically significant, and there is no significant difference between the two coefficients. Therefore, we conclude that a decrease in managers' incentives to mislead investors may not be a channel through which the TSPP reduces earnings guidance.

6. Further Analyses

To provide more in-depth analysis, we examine whether the TSPP's negative impact on earnings guidance varies with firms' proprietary costs of disclosure, with their future financing needs, and with their size. To conduct the cross-sectional analyses, we modify our baseline model in Equation (1) by adding the interaction terms among $Treat$, $Post$, and the cross-sectional (partitioning) measure. In addition, we investigate whether the effect of TSPP can be generalized to other forms of voluntary disclosures: firm-initiated press releases and voluntary 8-K disclosures.

6.1. Proprietary Cost of Disclosure

Economic theory predicts that corporate decisions, including those about disclosure, are based on trading off the costs and benefits of these decision outcomes. Heterogeneity in firms' disclosures occurs despite the documented benefits of transparency because different firms have different disclosure costs (Healy and Palepu 2001, Verrecchia 2001). Given this perspective, we posit that firms with higher disclosure costs are even more likely to forego disclosure when its benefits are reduced.

Prior studies (e.g., Verrecchia 1983) on earnings guidance document that proprietary costs, an important component of disclosure costs, are among the most important determinants of voluntary disclosure and may dissuade managers from issuing it. Prior evidence indicates that firms with more R&D investment have more proprietary information, meaning that their voluntary disclosures could have greater proprietary costs (Bamber and Cheon 1998, Jones 2007, Ellis et al. 2012, Merkley 2014, Cao et al. 2018, Kim and Valentine 2021, 2023). Accordingly, we use R&D expenditure as our first proxy for proprietary cost. R&D intensity ($R\&D\ Intensity$) is calculated as R&D expenses divided by the previous quarter's total assets. Then, we create a dummy variable, $High\ R\&D$, that equals one for firms with R&D intensity above the sample median in a quarter, and zero otherwise. Second, we follow Rawson (2022) and employ the state-level staggered adoption of the inevitable disclosure doctrine (IDD) as an exogenous shock that increases the marginal costs of proprietary information disclosure. IDD increases the proprietary cost of disclosure by restricting the movement of key employees, which limits the flow of vital information, thereby forcing other firms to rely more on other (e.g., public) sources of information disclosure (Li et al. 2018, Callen et al. 2020). We obtain the list of adopting states from Li et al. (2018) and generate IDD as a dummy

variable that equals one for firms headquartered in IDD-adopting states after the adoption, and zero otherwise.

In Table 7, columns (1) and (2), the coefficient on $Treat \times Post \times High\ R\&D$ is negative and marginally significant in column (1), though not significant in column (2). Economically, the results in column (1) suggest that when the dependent variable is $DumMF$, our findings imply that treatment firms with higher R&D intensity are 4.3% less likely to issue a guidance, compared with treatment firms with lower R&D intensity. The coefficient on $Treat \times Post \times IDD$ is negative and marginally significant in Table 7, column (4), but is not significant in column (3). In economic terms, the result in column (4) suggests that treatment firms headquartered in IDD-adopting states experience a 4.7% decrease in forecast frequency, compared with treatment firms in nonadopting states. Collectively, the findings are consistent with our expectation that managers of firms facing high disclosure costs have even less incentive to issue earnings guidance.

6.2. External Financing Needs

Although our results imply that firms reduce earnings guidance after TSPP initiation, we expect a heterogeneous reaction between firms facing greater versus weaker external financing need. The underlying intuition is that because voluntary disclosure affects the cost of external financing, a firm that cuts disclosure risks adverse impacts on its cost of capital (Francis et al. 2005, 2008; Cheynel 2013; Balakrishnan et al. 2014). The above studies suggest that the marginal cost of a decrease in earnings guidance might outweigh the marginal benefits for firms with high external financing needs. Accordingly, we expect the negative impact of the TSPP on earnings guidance to be less pronounced for firms with external financing needs.

We define firms with financing needs ($High\ Finance\ Needs$) using a dummy variable that equals one for firms that issue net equity or net debt in the next four quarters that is at least 5% of total assets, and zero otherwise.²¹ In Table 7, columns (5) and (6), the coefficients on $Treat \times Post \times High\ Finance\ Needs$ are significantly positive, indicating that the negative effect of the TSPP on earnings guidance is less pronounced for firms with future financing needs. Economically, when the dependent variable is $DumMF$ ($FreqMF$), compared with treatment firms with lower external financing needs, those with higher external financing needs experience a 4.4% (4.3%) lower reduction in guidance issuance (guidance frequency). The findings align with our expectation that the marginal cost of a decrease in earnings guidance outweighs the marginal benefits for firms facing greater external financing needs, making them less affected by the TSPP.

6.3. Firm Size

Because the TSPP mainly targets small-cap firms, firms in the program are typically smaller than those listed in major U.S. stock exchanges. Prior literature has documented that

Table 7. Cross-Sectional Analyses Based on Disclosure Costs, External Financing Needs, and Firm Size

	<i>DumMF_t</i>	<i>FreqMF_t</i>	<i>DumMF_t</i>	<i>FreqMF_t</i>	<i>DumMF_t</i>	<i>FreqMF_t</i>	<i>DumMF_t</i>	<i>FreqMF_t</i>
Cross-sectional test based on:	Disclosure costs				External financing needs		Firm size	
Moderating variable =	<i>High R&D</i>		<i>IDD</i>		<i>High Finance Needs</i>		<i>High Size</i>	
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Treat</i> × <i>Post</i> × <i>Moderating Variable</i>	−0.043* (−1.887)	−0.029 (−1.318)	−0.031 (−1.221)	−0.047* (−1.938)	0.044** (1.966)	0.043** (2.100)	0.010 (0.432)	0.014 (0.658)
<i>Treat</i> × <i>Post</i>	−0.007 (−0.480)	−0.009 (−0.652)	−0.013 (−0.643)	0.002 (0.107)	−0.050*** (−2.899)	−0.046*** (−2.782)	−0.032** (−2.267)	−0.029** (−2.294)
<i>Treat</i> × <i>Moderating Variable</i>	0.038 (1.475)	0.036 (1.500)	−0.168 (−0.559)	−0.103 (−0.474)	−0.027 (−1.447)	−0.021 (−1.161)	0.055 (1.175)	0.065 (1.499)
<i>Post</i> × <i>Moderating Variable</i>	0.033** (2.058)	0.027* (1.857)	0.023 (1.270)	0.036** (2.086)	−0.013 (−0.776)	−0.013 (−0.853)	−0.001 (−0.080)	−0.015 (−1.023)
<i>Moderating Variable</i>	0.012 (0.252)	−0.033 (−0.737)	−0.168 (−1.520)	−0.138 (−1.430)	0.030 (0.748)	0.040 (1.075)	0.022 (0.349)	0.021 (0.373)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Moderating Variable</i> × Controls	Yes	Yes	No	No	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	14,157	14,157	11,863	11,863	14,149	14,149	14,157	14,157
Adjusted R ²	0.831	0.801	0.836	0.803	0.830	0.801	0.831	0.801

Notes. This table presents results of cross-sectional variations in the documented relation between the Tick Size Pilot Program and earnings guidance. Results in columns (1)–(4) are based on the cost of disclosure, columns (5) and (6) are based on external financing needs, and columns (7) and (8) are based on firm size. The regressions are performed using OLS. The *t*-statistics in parentheses are adjusted for heteroscedasticity and clustered by firm. The appendix presents the variable definitions.

p* < 0.10; *p* < 0.05; ****p* < 0.01.

large firms are more likely to provide management guidance (Cox 1985, Chow and Wong-Boren 1987, Kasznik and Lev 1995). From the disclosure-demand perspective, large firms are more likely than small firms to face investor attention and, thus, pressure to provide earnings guidance.²² From the disclosure supply perspective, larger firms are likely to have lower costs (e.g., proprietary cost) of providing earnings guidance, compared with smaller firms. It is ex ante unclear how the sample of relatively small-sized firms with lower propensity of offering guidance in our analysis would affect the generalizability of our primary finding that an increase in tick size of a firm's stock reduces its issuance of earnings guidance.

To address this issue, we test the moderating role of firm size on the documented relation between the TSPP and earnings guidance. We define *High Size* as a dummy variable that equals one for firms with above-median firm size in a year-quarter, and zero otherwise. In Table 7, columns (7) and (8), the coefficients on *Treat* × *Post* × *High Size* are not statistically significant, suggesting that our results may be generalizable to large firms.

6.4. Alternative Measures of Voluntary Disclosure

In the last section, we consider firm-initiated press releases and voluntary 8-K disclosures as alternative measures of voluntary disclosure. We collect firm-initiated press-release data from the RavenPack database and 8-K disclosure data from EDGAR. Press-release likelihood (*DumPR*) is a dummy variable that equals one if the firm issues at least one press release during the current quarter, and zero

otherwise. Press-release frequency (*FreqPR*) is the natural logarithm of one plus the number of press releases that the firm issues in the current quarter. The 8-K likelihood (*Dum8K*) is a dummy variable that equals one if the firm issues at least one voluntary 8-K disclosure during the current quarter, and zero otherwise. The 8-K frequency (*Freq8K*) is the natural logarithm of one plus the number of voluntary 8-K disclosures that the firm issues in the current quarter. We identify voluntary 8-K disclosures using the approach in He and Plumlee (2020).²³

Next, we re-estimate the regression in Equation (1) by replacing the earnings guidance variables (i.e., *DumMF* and *FreqMF*) with the four variables. The results are presented in Table 8. In columns (1) and (2), the coefficients on *Treat* × *Post* are negative and significant, indicating fewer press releases for the treatment firms, relative to the control firms, after initiation of the TSPP. The coefficients on *Treat* × *Post* are not significant in columns (3) and (4), implying that the TSPP does not affect firms' voluntary 8-K disclosures.²⁴ Overall, these results show that the TSPP also affects other forms of voluntary disclosure.

7. Conclusion

How firms strategically adjust their earnings guidance in response to changes in stock market conditions is an important question, especially considering that earnings guidance primarily serves to reduce information asymmetry between firms and their investors. In this study, we use the 2016 TSPP as an exogenous shock to examine the relation between an increase in tick size and earnings guidance.

Table 8. Alternative Measures of Voluntary Disclosure

Variable	<i>DumPR_t</i> (1)	<i>FreqPR_t</i> (2)	<i>Dum8K_t</i> (3)	<i>Freq8K_t</i> (4)
<i>Treat</i> × <i>Post</i>	−0.043*** (−3.336)	−0.063** (−2.474)	−0.021 (−1.291)	0.003 (0.215)
<i>Size_{t−1}</i>	−0.012 (−0.620)	0.009 (0.267)	−0.009 (−0.493)	−0.083*** (−4.562)
<i>BM_{t−1}</i>	0.003 (0.167)	−0.040 (−1.374)	−0.024 (−1.165)	−0.047*** (−2.608)
<i>ROA_{t−1}</i>	−0.017 (−0.159)	−0.241 (−1.351)	0.126 (1.021)	−0.181 (−1.572)
<i>Loss_{t−1}</i>	−0.009 (−0.843)	−0.010 (−0.537)	−0.016 (−1.203)	−0.004 (−0.374)
<i>Lev_{t−1}</i>	−0.083* (−1.666)	−0.159* (−1.685)	0.022 (0.412)	−0.152*** (−3.025)
<i>IO_{t−1}</i>	−0.023 (−0.391)	0.002 (0.023)	−0.021 (−0.413)	0.108* (1.711)
<i>AnalystCov_{t−1}</i>	−0.009 (−0.644)	0.016 (0.620)	−0.007 (−0.431)	−0.019 (−1.160)
<i>StkVol_{t−1}</i>	−0.243* (−1.755)	−0.436* (−1.742)	−0.193 (−1.057)	−0.045 (−0.316)
<i>EarnVol_{t−1}</i>	−0.265*** (−2.692)	−0.258 (−1.411)	−0.040 (−0.311)	−0.110 (−0.983)
<i>Ret_{t−1}</i>	−0.010 (−0.170)	−0.013 (−0.134)	0.041 (0.528)	−0.026 (−0.455)
<i>Turnover_{t−1}</i>	0.006 (0.468)	0.006 (0.252)	0.005 (0.347)	0.018 (1.315)
<i>Cash_{t−1}</i>	−0.017 (−0.309)	−0.050 (−0.472)	0.050 (0.830)	−0.031 (−0.501)
<i>Tangibility_{t−1}</i>	−0.014 (−0.142)	0.038 (0.228)	0.053 (0.396)	0.176 (1.575)
<i>EquityIssue_{t−1}</i>	0.008 (0.757)	0.011 (0.621)	0.008 (0.633)	0.006 (0.584)
<i>DebtIssue_{t−1}</i>	−0.009 (−0.999)	−0.028* (−1.725)	−0.006 (−0.504)	0.020** (2.394)
<i>Litigation_{t−1}</i>	−0.051* (−1.905)	−0.094** (−2.006)	0.009 (0.265)	0.028 (1.113)
<i>Sync_t</i>	−0.002 (−0.634)	−0.005 (−0.939)	0.001 (0.287)	−0.007** (−2.064)
<i>Illiquidity_t</i>	−0.000 (−0.378)	0.000 (0.334)	0.000 (0.493)	0.001 (1.081)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes
N	14,157	14,157	14,157	14,157
Adjusted R ²	0.471	0.450	0.056	0.575

Notes. This table presents the results of the effect of the Tick Size Pilot Program on the likelihood and frequency of firm-initiated voluntary press releases and voluntary disclosures in 8-K filings. The regressions are performed using OLS. The *t*-statistics in parentheses are adjusted for heteroscedasticity and clustered by firm. The appendix presents the variable definitions.

p* < 0.10; *p* < 0.05; ****p* < 0.01.

Building upon recent literature documenting that the TSPP encouraged investors to engage in fundamental information acquisition and led to improvements in financial reporting quality, we study the effect of the program on earnings guidance. We find that the TSPP reduced treatment firms' propensity to issue earnings guidance, consistent with firms reducing potentially costly disclosures when the benefit of lessening information asymmetry between firms and their investors is lower.

We also show that the reduction in earnings guidance for treatment firms emerged in the two quarters following the TSPP's initiation, confirming that the parallel-trend assumption behind the difference-in-differences model holds in our setting. In additional analyses, we confirm that the negative effect of the TSPP on earnings guidance is concentrated among treatment firms experiencing increases in fundamental information acquisition and improvements in financial reporting quality, consistent with these two changes serving as mechanisms through which increased tick size reduces earnings guidance. We perform two cross-sectional analyses to further understand the relation between the TSPP and earnings guidance. We find that the negative effect of the TSPP on earnings guidance is stronger for firms with higher disclosure costs and weaker for firms with greater future financing needs, consistent with the notion that firms adjust their voluntary disclosure in response to an increase in tick size based on the benefit-cost trade-off of reducing earnings guidance.

One limitation of our study is that our inferences are limited to small-cap firms because the TSPP targeted small-cap firms, which are less likely than large-cap firms to provide earnings guidance. Although our arguments regarding decreased earnings guidance after an increase in tick size are likely to apply to large-cap firms, we have no direct way of confirming this, given the nature of the TSPP. However, we address the issue of generalizability indirectly by showing a negative effect of increased tick size for both relatively larger-cap and smaller-cap firms in our sample. We acknowledge that the concern about generalizability remains.

The growing literature examining various outcomes of the TSPP underscores the importance of understanding the impact of tick size and the utility of this program as a natural experiment that helps identify causal effects. Our paper generally extends the literature that uses fundamental information acquisition as the channel to examine further outcomes of the TSPP (e.g., Ahmed et al. 2020, Ye et al. 2023). However, it is possible that other channels (e.g., liquidity) could link tick size to earnings guidance (Hope and Liu 2022). Even if we provide evidence that fundamental information acquisition and the associated improvements in financial reporting quality are the mechanisms through which the TSPP affects earnings guidance, we acknowledge that other channels may explain our baseline findings.

Acknowledgments

The authors appreciate the helpful comments of the Department Editor (Suraj Srinivasan), the Associate Editor, two anonymous reviewers, and seminar and conference participants at the Hong Kong Polytechnic University, Central University of Finance and Economics, Renmin University of China, and the 2020 Miami Rookie Camp. This paper was previously titled, "Informed Investors and Earnings Guidance: Evidence from SEC Tick Size Pilot Program."

Appendix. Variable Definitions

Variable	Definition
Variables in the baseline analysis in Table 3	
<i>DumMF</i>	Dummy variable that equals one if the firm issues at least one earnings guidance during the current quarter, and zero otherwise.
<i>FreqMF</i>	Natural logarithm of one plus the number of earnings guidance the firm issues during the current quarter.
<i>Treat</i>	Dummy variable that equals one if the firm belongs to the treatment group, and zero if it belongs to the control group.
<i>Post</i>	Dummy variable that equals one for the postpilot period (i.e., January 1, 2017, to September 30, 2018), and zero for the prepilot period (i.e., January 1, 2015, to September 31, 2016).
<i>Size</i>	Natural logarithm of total assets.
<i>BM</i>	Book value of equity divided by its market value.
<i>ROA</i>	Income before extraordinary items divided by total assets.
<i>Loss</i>	Dummy variable that equals one if income before extraordinary items is negative, and zero otherwise.
<i>Lev</i>	Long-term debt divided by total assets.
<i>IO</i>	Percentage of shares held by institutional investors.
<i>AnalystCov</i>	Natural logarithm of one plus the number of analysts covering the firm in the current quarter.
<i>StkVol</i>	Standard deviation of the daily stock return in the current quarter.
<i>EarnVol</i>	Standard deviation of the quarterly return on assets over the past four quarters.
<i>Ret</i>	Buy-and-hold size-adjusted return over the current quarter.
<i>Turnover</i>	Average monthly share turnover during the current quarter, where monthly share turnover is the monthly trading volume divided by the total number of shares outstanding at month-end.
<i>Cash</i>	Cash and cash equivalents divided by total assets.
<i>Tangibility</i>	Net property, plant, and equipment divided by total assets.
<i>EquityIssue</i>	Dummy variable that equals one if the firm issues equity during the quarter, and zero otherwise.
<i>DebtIssue</i>	Dummy variable that equals one if the firm issues debt during the quarter, and zero otherwise.
<i>Litigation</i>	Ex ante class action litigation risk, calculated using the coefficient from model (3) in Kim and Skinner (2012).
<i>Sync</i>	The natural logarithm of R^2 divided by $1 - R^2$, where R^2 is estimated from a quarterly time-series regression of weekly stock returns using Carhart's (1997) four-factor model.
<i>Illiquidity</i>	Amihud's (2002) illiquidity ratio, defined as the natural logarithm of one plus the quarterly mean of Amihud's (2002) daily ratio.
Additional variables in Table 4	
<i>Pre6</i>	Dummy variable that equals one if the quarter is the second quarter of 2015, and zero otherwise.
<i>Pre5</i>	Dummy variable that equals one if the quarter is the third quarter of 2015, and zero otherwise.
<i>Pre4</i>	Dummy variable that equals one if the quarter is the fourth quarter of 2015, and zero otherwise.
<i>Pre3</i>	Dummy variable that equals one if the quarter is the first quarter of 2016, and zero otherwise.
<i>Pre2</i>	Dummy variable that equals one if the quarter is the second quarter of 2016, and zero otherwise.
<i>Pre1</i>	Dummy variable that equals one if the quarter is the third quarter of 2016, and zero otherwise.
<i>Post1</i>	Dummy variable that equals one if the quarter is the first quarter of 2017, and zero otherwise.
<i>Post2</i>	Dummy variable that equals one if the quarter is the second quarter of 2017, and zero otherwise.
<i>Post3</i>	Dummy variable that equals one if the quarter is the third quarter of 2017, and zero otherwise.
<i>Post4</i>	Dummy variable that equals one if the quarter is the fourth quarter of 2017, and zero otherwise.
<i>Post5</i>	Dummy variable that equals one if the quarter is the first quarter of 2018, and zero otherwise.
<i>Post6</i>	Dummy variable that equals one if the quarter is the second quarter of 2018, and zero otherwise.
<i>Post7</i>	Dummy variable that equals one if the quarter is the third quarter of 2018, and zero otherwise.
Additional variables in Table 5	
<i>Pseudo_Treat</i>	Dummy variable that equals one if the firm belongs to the pseudo-treatment group, and zero if it belongs to the pseudo-control group.
<i>Pseudo_Post</i>	Dummy variable that equals one for the pseudo-postpilot period (i.e., April 1, 2015, to December 31, 2016), and zero for the pseudo-prepilot period (i.e., July 1, 2013, to March 31, 2015).
Additional variables in Table 6	
<i>Treat_FIAIncMore</i>	Dummy variable that equals one for treatment firms that experienced a change in average fundamental information acquisition above the sample median value of all treatment firms, and zero otherwise.
<i>Treat_FIAIncLess</i>	Dummy variable that equals one for treatment firms that experienced a change in average fundamental information acquisition below the sample median value of all treatment firms, and zero otherwise.
<i>Treat_FRQualityIncMore</i>	Dummy variable that equals one for the treatment firms that experienced a change in average financial reporting quality above the sample median of all treatment firms, and zero otherwise.
<i>Treat_FRQualityIncLess</i>	Dummy variable that equals one for the treatment firms that experienced a change in average financial reporting quality below the sample median of all treatment firms, and zero otherwise.

Appendix. (Continued)

Variable	Definition
<i>Treat_HighEMI</i>	Dummy variable that equals one for treatment firms with average expectations management above the sample median of all treatment firms, and zero otherwise.
<i>Treat_LowEMI</i>	Dummy variable that equals one for treatment firms with average expectations management below the sample median of all treatment firms, and zero otherwise.
Additional variables in Table 7	
<i>High R&D</i>	Dummy variable that equals one for firms with R&D intensity above the sample median in a quarter, and zero otherwise.
<i>IDD</i>	Dummy variable that equals one for firms headquartered in a state that has adopted the inevitable disclosure doctrine, and zero otherwise.
<i>High Finance Needs</i>	Dummy variable that equals one for firms that issue net equity or net debt in the next four quarters that is at least 5% of total assets, and zero otherwise.
<i>High Size</i>	Dummy variable that equals one for firms with above-median firm size in a quarter, and zero otherwise.
Additional variables in Table 8	
<i>DumPR</i>	Dummy variable that equals one for firms that issue at least one press release in the current quarter, and zero otherwise.
<i>FreqPR</i>	Natural logarithm of one plus the number of firm initiated press releases in the current quarter.
<i>Dum8K</i>	Dummy variable that equals one for firms that issue at least one voluntary 8-K disclosure during the current quarter, and zero otherwise.
<i>Freq8K</i>	Natural logarithm of one plus the number of voluntary 8-K disclosure the firm issues in the current quarter.

Endnotes

- ¹ To the best of our knowledge, despite an extensive literature assuming or arguing that voluntary disclosure is provided to reduce information asymmetry between the firm and its investors, prior research has not examined how firms respond in terms of voluntary disclosure when investors become more informed, likely due to a lack of a feasible identification of exogenous conditions under which investors become more informed.
- ² For example, a recent study by Al Guindy et al. (2020) finds that the broad and nonexclusionary disclosure requirements of RegFD inhibit adoption of new disclosure technologies.
- ³ The basic argument in Hope and Liu (2022) is that tick size increases induced by the TSPP lead to decreased stock liquidity, which, in turn, reduces stock price reactions to earnings guidance. As managers regard this reduced stock price reaction as a reduction in the benefit of providing guidance, they provide less frequent earnings guidance. However, it is unclear whether an unambiguous decrease in liquidity occurs after the tick size increase. For example, Chung et al. (2020) show that although liquidity for small orders decreases, liquidity for large orders actually improves for treatment firms. Nevertheless, we control for stock liquidity in all our estimations.
- ⁴ See the Order Directing the Exchanges and the Financial Industry Regulatory Authority to Submit a Tick Size Pilot Plan (<http://www.sec.gov/rules/other/2014/34-72460.pdf>).
- ⁵ See the Plan to Implement a Tick Size Pilot Program (<https://www.sec.gov/divisions/marketreg/tick-size-pilot-plan-final.pdf>) and the SEC’s Statement on the Expiration of the Tick Size Pilot (<https://www.sec.gov/news/public-statement/tm-dera-expiration-tick-size-pilot>).
- ⁶ Under the trade-at requirement, a trading center could not file an incoming order without displaying the best price. This requirement discouraged order flows from being channeled to dark or alternative trading venues (Griffith and Roseman 2019). It also prevented nondisplayed orders from trading at the bid or offer, except for certain exemptions (i.e., retail orders had to be executed with a minimum \$0.0005 price improvement and a block size order greater than 5,000 shares or \$100,000).
- ⁷ Undercutting, or frontrunning, occurs when a broker or trader attempts to step in front of other limit orders. As major frontrunners, algorithmic traders trade frequently and can gain price priority to undercut limit orders at the newly quoted spread. A larger tick size increases the frontrunners’ costs and reduces the risk of undercutting (e.g., Goettler et al. 2005, Rindi and Werner 2019).
- ⁸ With regard to the costs of providing guidance, disclosing forward-looking information may increase proprietary costs that can damage firms’ competitive positions (Verrecchia 1983, Huang et al. 2017). We examine these costs in Section 5.2.
- ⁹ Indeed, Ye et al. (2023) find that a larger tick size increases a firm’s investment sensitivity to stock prices, suggesting that managers glean more new information from stock prices to guide their investment decisions as tick size increases.
- ¹⁰ Because of the formulation of our primary hypothesis as a null hypothesis, we develop cross-sectional analyses to provide more in-depth insights after examining the initial empirical results for this hypothesis.
- ¹¹ See the Tick Size Pilot Data Collection Securities Files, available at <https://www.finra.org/rules-guidance/key-topics/tick-size-pilot-program/data-collection-securities-and-pilot-securities-files>.
- ¹² Following prior studies (e.g., Ajinkya et al. 2005, Houston et al. 2010, Kim et al. 2018), we exclude pre-announcements because such forecasts are considered part of an earnings announcement strategy, rather than a voluntary disclosure activity.
- ¹³ We use an OLS regression for *DumMF* in the main regression because we include firm fixed effects. When the number of fixed effects is large, a logit regression is inconsistent and suffers from quasiseperation issues (Albert and Anderson 1984). Also, the inclusion of firm fixed effects in a logit model typically results in a significant drop in the number of observations.
- ¹⁴ Although Ahmed et al. (2020) show that the TSPP leads to improved financial reporting quality, we do not control for financial reporting quality because we argue that it is one potential mechanism through which the TSPP could affect earnings guidance.
- ¹⁵ The economic magnitude is meaningful and comparable to prior studies that examine the impact of firm information environment on

earnings guidance (e.g., Chen et al. 2008, Anantharaman and Zhang 2011). For instance, Chen et al. (2008) find that due to lower information asymmetry between managers and owners, family firms are less likely than nonfamily firms to issue management forecasts. The authors document an economic magnitude of roughly 14%. Anantharaman and Zhang (2011) find that an exogenous decrease in analyst coverage increases the likelihood of management guidance, and they document an economic magnitude of about 29%. Given that the TSPP was not targeted to influence corporate disclosures, an economic magnitude of about 11% seems reasonable.

¹⁶ This delay might be because firms need time to recognize changes in investor fundamental information acquisition and financial reporting quality and to adjust their earnings guidance strategies accordingly. Ahmed et al. (2020) also find that the increase in financial reporting quality for treatment firms emerges three quarters after initiation of the pilot program (i.e., the second half of 2017), though an increase in tick size after initiation of the TSPP may have an immediate effect on investors' information acquisition.

¹⁷ Given that we document a reduction in guidance, one might question whether it would be more appropriate to restrict our sample for the main analysis to only firms that issue at least one earnings guidance, as these firms are the only type that can reduce guidance. However, we develop our primary hypothesis as a null hypothesis due to the tension in the arguments about whether the TSPP would lead to an increase or decrease in guidance. Thus, from an ex ante perspective (i.e., before observing the results), we believe it is appropriate to use all firms for our main analysis.

¹⁸ All our results and inferences remain similar when we subject our analyses to the sample with at least one earnings guidance over our sample period. Also, it is worth noting that the coefficients in this test are significantly larger in magnitude than those in columns (2) and (4) of Table 3 (the coefficients in this test are -0.078 and -0.064 , with the corresponding ones in columns (2) and (4) of Table 3 being -0.027 and -0.022 , respectively). One possible explanation is that only firms that have issued guidance can reduce their guidance; thus, including firms that cannot reduce guidance leads to the documentation of a smaller average reduction in guidance.

¹⁹ Data for the fundamental information-acquisition measures are obtained from Dr. James Ryans' website at <http://www.jamesryans.com>. See Ryans (2017) for a detailed explanation of the methodology used in constructing these measures.

²⁰ Because the data on the three fundamental information-acquisition measures are available only until the second quarter of 2017, we use data on the first two quarters of 2017 to calculate average fundamental information acquisition in the post-TSPP period (Ahmed et al. 2020, Ye et al. 2023).

²¹ The 5% cut-off follows prior studies (e.g., Chang et al. 2006, 2009). One advantage of using a dummy variable to capture external financing needs is its ability to isolate firms that issue (or need) substantial amounts of debt or equity (or both) for major corporate investments (Korajczyk and Levy 2003; Chang et al. 2006, 2009) and firms for which voluntary disclosure matters most.

²² One might also argue that there is greater information asymmetry between small firms and their investors, and, thus, these firms face greater disclosure demand (Bushee et al. 2010). However, evidence in prior management guidance literature generally does not support this argument.

²³ Following He and Plumlee (2020), we classify only three 8-K items as voluntary: item 2.02 ("Results of Operations and Financial Condition"), item 7.01 ("Regulation FD Disclosure"), and item 8.01 ("Other Events").

²⁴ The insignificant results for voluntary 8-K disclosure are likely because 8-K disclosures are event-based, and only a small proportion of 8-K items are voluntary. The SEC requires firms to promptly

disclose information about material events through 8-K filings, and among the 22 items, only three are voluntary or semivoluntary (He and Plumlee 2020), which further limits managers' discretion in voluntarily disclosing information in 8-K filings.

References

- Ahmed AS, Li Y, Xu N (2020) Tick size and financial reporting quality in small-cap firms: Evidence from a natural experiment. *J. Accounting Res.* 58(4):869–914.
- Ajinkya B, Bhojraj S, Sengupta P (2005) The association between outside directors, institutional investors and the properties of management earnings forecasts. *J. Accounting Res.* 43(3):343–376.
- Al Guindy MA, Naughton JP, Riordan R (2020) The chilling effect of regulation FD: Evidence from the SEC endorsement of social media. Preprint, submitted April 15, <https://dx.doi.org/10.2139/ssrn.3385180>.
- Albert A, Anderson JA (1984) On the existence of maximum likelihood estimates in logistic regression models. *Biometrika* 71(1):1–10.
- Albuquerque R, Song S, Yao C (2020) The price effects of liquidity shocks: A study of the SEC's tick size experiment. *J. Financial Econom.* 138(3):700–724.
- Amihud Y (2002) Illiquidity and stock returns: Cross-section and time-series effects. *J. Financial Markets* 5(1):31–56.
- Anantharaman D, Zhang Y (2011) Cover me: Managers' responses to changes in analyst coverage in the post-regulation FD period. *Accounting Rev.* 86(6):1851–1885.
- Balakrishnan K, Billings M, Kelly B, Ljungqvist A (2014) Shaping liquidity: On the causal effects of voluntary disclosure. *J. Finance* 69(5):2237–2278.
- Bamber LS, Cheon YS (1998) Discretionary management earnings forecast disclosures: Antecedents and outcomes associated with forecast venue and forecast specificity choices. *J. Accounting Res.* 36(2):167–190.
- Beyer A, Cohen DA, Lys TZ, Walther BR (2010) The financial reporting environment: Review of the recent literature. *J. Accounting Econom.* 50(2–3):296–343.
- Blankespoor E, deHaan E, Marinovic I (2020) Disclosure processing costs, investors' information choice, and equity market outcomes: A review. *J. Accounting Econom.* 70(2–3):101344.
- Bond P, Edmans A, Goldstein I (2012) The real effects of financial markets. *Annu. Rev. Financial Econom.* 4(1):339–360.
- Bourveau T, Lou Y, Wang R (2018) Shareholder litigation and corporate disclosure: Evidence from derivative lawsuits. *J. Accounting Res.* 56(3):797–842.
- Bushee BJ, Core JE, Guay W, Hamm SJ (2010) The role of the business press as an information intermediary. *J. Accounting Res.* 48(1):1–19.
- Carhart MM (1997) On persistence in mutual fund performance. *J. Finance* 52(1):57–82.
- Callen JL, Fang X, Zhang W (2020) Protection of proprietary information and financial reporting opacity: Evidence from a natural experiment. *J. Corporate Finance* 64:101641.
- Cao SS, Ma G, Tucker JW, Wan C (2018) Technological peer pressure and product disclosure. *Accounting Rev.* 93(6):95–126.
- Chang X, Dasgupta S, Hilary G (2006) Analyst coverage and financing decisions. *J. Finance* 61(6):3009–3048.
- Chang X, Dasgupta S, Hilary G (2009) The effect of auditor quality on financing decisions. *Accounting Rev.* 84(4):1085–1117.
- Chen S, Chen X, Cheng Q (2008) Do family firms provide more or less voluntary disclosure? *J. Accounting Res.* 46(3):499–536.
- Chen Y, Ng J, Yang X (2021) Talk less, learn more: Strategic disclosure in response to managerial learning from the options market. *J. Accounting Res.* 59(5):1609–1649.
- Cheyne E (2013) A theory of voluntary disclosure and cost of capital. *Rev. Accounting Stud.* 18(4):987–1020.
- Chow CW, Wong-Boren A (1987) Voluntary financial disclosure by Mexican corporations. *Accounting Rev.* 62(3):533–541.

- Chung KH, Lee AJ, Rösch D (2020) Tick size, liquidity for small and large orders, and price informativeness: Evidence from the Tick Size Pilot Program. *J. Financial Econom.* 136(3):879–899.
- Coller M, Yohn TL (1997) Management forecasts and information asymmetry: An examination of bid-ask spreads. *J. Accounting Res.* 35(2):181–191.
- Collins DW, Pungaliya RS, Vijh AM (2017) The effects of firm growth and model specification choices on tests of earnings management in quarterly settings. *Accounting Rev.* 92(2):69–100.
- Cotter J, Tuna I, Wysocki PD (2006) Expectations management and beatable targets: How do analysts react to explicit earnings guidance? *Contemp. Accounting Res.* 23(3):593–624.
- Cox C (1985) Further evidence on the representativeness of management earnings forecasts. *Accounting Rev.* 60:692–701.
- Dechow PM, Sloan RG, Sweeney AP (1995) Detecting earnings management. *Accounting Rev.* 70(2):193–225.
- Diamond DW, Verrecchia RE (1991) Disclosure, liquidity, and the cost of capital. *J. Finance* 46(4):1325–1359.
- Drake MS, Roulstone DT, Thornock JR (2015) The determinants and consequences of information acquisition via EDGAR. *Contemp. Accounting Res.* 32(3):1128–1161.
- Ellis JA, Fee CE, Thomas SE (2012) Proprietary costs and the disclosure of information about customers. *J. Accounting Res.* 50(3):685–727.
- Francis J, Nanda D, Olsson P (2008) Voluntary disclosure, earnings quality, and cost of capital. *J. Accounting Res.* 46(1):53–99.
- Francis JR, Khurana IK, Pereira R (2005) Disclosure incentives and effects on cost of capital around the world. *Accounting Rev.* 80(4):1125–1162.
- Gao P, Liang PJ (2013) Informational feedback, adverse selection, and optimal disclosure policy. *J. Accounting Res.* 51(5):1133–1158.
- Glaeser S (2018) The effects of proprietary information on corporate disclosure and transparency: Evidence from trade secrets. *J. Accounting Econom.* 66(1):163–193.
- Goettler RL, Parlour CA, Rajan U (2005) Equilibrium in a dynamic limit order market. *J. Finance* 60(5):2149–2192.
- Griffith TG, Roseman BS (2019) Making cents of tick sizes: The effect of the 2016 US SEC tick size pilot on limit order book liquidity. *J. Banking Finance* 101:104–121.
- He J, Plumlee M (2020) Measuring disclosure using 8-K filings. *Rev. Accounting Stud.* 25(3):903–962.
- Healy PM, Palepu KG (2001) Information asymmetry, corporate disclosure, and the capital markets: A review of the empirical disclosure literature. *J. Accounting Econom.* 31(1–3):405–440.
- Hope OK, Liu J (2022) Does stock liquidity shape voluntary disclosure? Evidence from the SEC tick size pilot program. *Rev. Accounting Stud.*, ePub ahead of print April 18, <https://doi.org/10.1007/s11142-022-09686-0>.
- Houston JF, Lev B, Tucker JW (2010) To guide or not to guide? Causes and consequences of stopping quarterly earnings guidance. *Contemp. Account. Res.* 27(1):143–185.
- Huang S, Ng J, Ranasinghe T, Zhang M (2021) Do innovative firms communicate more? Evidence from the relation between patenting and management guidance. *Accounting Rev.* 96(1):273–297.
- Huang Y, Jennings R, Yu Y (2017) Product market competition and managerial disclosure of earnings forecasts: Evidence from import tariff rate reductions. *Accounting Rev.* 92(3):185–207.
- Hutton AP, Lee LF, Shu SZ (2012) Do managers always know better? The relative accuracy of management and analyst forecasts. *J. Accounting Res.* 50(5):1217–1244.
- Jayaraman S, Wu JS (2020) Should I stay or should I grow? Using voluntary disclosure to elicit market feedback. *Rev. Financial Stud.* 33(8):3854–3888.
- Johnson TL, Kim J, So EC (2020) Expectations management and stock returns. *Rev. Financial Stud.* 33(10):4580–4626.
- Jones DA (2007) Voluntary disclosure in R&D-intensive industries. *Contemp. Accounting Res.* 24(2):489–522.
- Jones JJ (1991) Earnings management during import relief investigations. *J. Accounting Res.* 29:193–228.
- Kasznik R, Lev B (1995) To warn or not to warn: Management disclosures in the face of an earnings surprise. *Accounting Rev.* 70:113–134.
- Kim I, Skinner DJ (2012) Measuring securities litigation risk. *J. Accounting Econom.* 53(1–2):290–310.
- Kim J, Valentine K (2021) The innovation consequences of mandatory patent disclosures. *J. Accounting Econom.* 71(2–3):101381.
- Kim J, Valentine K (2023) Public firm disclosures and the market for innovation. *J. Accounting Econom.* 76(1):101577.
- Kim JB, Shroff P, Vyas D, Wittenberg-Moerman R (2018) Credit default swaps and managers' voluntary disclosure. *J. Accounting Res.* 56(3):953–988.
- Koch AS, Lefanowicz CE, Robinson JR (2013) Regulation FD: A review and synthesis of the academic literature. *Accounting Horizons* 27(3):619–646.
- Korajczyk RA, Levy A (2003) Capital structure choice: Macroeconomic conditions and financial constraints. *J. Financial Econom.* 68(1):75–109.
- Kothari SP (2001) Capital markets research in accounting. *J. Accounting Econom.* 31(1–3):105–231.
- Kothari SP, Leone AJ, Wasley CE (2005) Performance matched discretionary accrual measures. *J. Accounting Econom.* 39(1):163–197.
- Kothari SP, Li X, Short JE (2009) The effect of disclosures by management, analysts, and business press on cost of capital, return volatility, and analyst forecasts: A study using content analysis. *Accounting Rev.* 84(5):1639–1670.
- Lee C, Watts E (2021) Tick size tolls: Can a trading slowdown improve earnings news discovery? *Accounting Rev.* 96(3):373–401.
- Leuz C, Wysocki PD (2016) The economics of disclosure and financial reporting regulation: Evidence and suggestions for future research. *J. Accounting Res.* 54(2):525–622.
- Li X, Yang HI (2016) Mandatory financial reporting and voluntary disclosure: The effect of mandatory IFRS adoption on management forecasts. *Accounting Rev.* 91(3):933–953.
- Li Y, Lin Y, Zhang L (2018) Trade secrets law and corporate disclosure: Causal evidence on the proprietary cost hypothesis. *J. Accounting Res.* 56(1):265–308.
- Loughran T, McDonald B (2017) The use of EDGAR filings by investors. *J. Behav. Finance* 18(2):231–248.
- Maslar DA, Serfling M, Shaikh S (2021) Economic downturns and the informativeness of management earnings guidance. *J. Accounting Res.* 59(4):1481–1520.
- Massa M, Zhang B, Zhang H (2015) The invisible hand of short selling: Does short selling discipline earnings management? *Rev. Financial Stud.* 28(6):1701–1736.
- Matsumoto DA (2002) Management's incentives to avoid negative earnings surprises. *Accounting Rev.* 77(3):483–514.
- Merkley KJ (2014) Narrative disclosure and earnings performance: Evidence from R&D disclosures. *Accounting Rev.* 89(2):725–757.
- Nagar V, Schoenfeld J, Wellman L (2019) The effect of economic policy uncertainty on investor information asymmetry and management disclosures. *J. Accounting Econom.* 67(1):36–57.
- Noh S, So EC, Weber JP (2019) Voluntary and mandatory disclosures: Do managers view them as substitutes? *J. Accounting Econom.* 68(1):101243.
- Patell JM (1976) Corporate forecasts of earnings per share and stock price behavior: Empirical test. *J. Accounting Res.* 14(2):246–276.
- Rawson C (2022) Manager perception and proprietary investment disclosure. *Rev. Accounting Stud.* 27:1493–1525.
- Rindi B, Werner I (2019) US tick size pilot. Working paper, Ohio State University, Columbus, OH.
- Rogers J, Stocken P (2005) Credibility of management forecasts. *Accounting Rev.* 80(4):1233–1260.
- Rogers JL, Van Buskirk A (2013) Bundled forecasts in empirical accounting research. *J. Accounting Econom.* 55(1):43–65.

- Ryans JP (2017) Using the EDGAR log file data set. Preprint, submitted February 8, <https://dx.doi.org/10.2139/ssrn.2913612>.
- Ryans JP (2021) Textual classification of SEC comment letters. *Rev. Accounting Stud.* 26(1):37–80.
- Stiglitz JE (2014) Tapping the brakes: Are less active markets safer and better for the economy? <https://www.frbatlanta.org/-/media/documents/news/conferences/2014/fmc/Stiglitz.pdf>.
- Verrecchia RE (1983) Discretionary disclosure. *J. Accounting Econom.* 5:179–194.
- Verrecchia RE (2001) Essays on disclosure. *J. Accounting Econom.* 32(1–3):97–180.
- Weller BM (2018) Does algorithmic trading reduce information acquisition? *Rev. Financial Stud.* 31(6):2184–2226.
- Ye M, Zheng MY, Zhu W (2023) The effect of tick size on managerial learning from stock prices. *J. Accounting Econom.* 75(1): 101515.
- Zuo L (2016) The informational feedback effect of stock prices on management forecasts. *J. Accounting Econom.* 61(2–3):391–413.